Estimation and inference in penalized function-on-function regression with within-curve residual dependence

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Penalized function-on-function regression, as implemented in the R package refund, considers all grid values as independent conditional on the linear predictor, but some dependence typically remains. We study the consequences for estimates and pointwise intervals in a simulation and a plasmode study on nine real datasets. Under dependence, restricted maximum likelihood (REML) undersmooths the coefficient surface: in our simulations' main settings, its relative error is two to four times that of curve-blocked neighborhood cross-validation (NCV). In our plasmode study, both fits have large errors, possibly through weak identification, and NCV is usually, but not always, better. Under dependence, the independence-model-based intervals for means and surfaces undercover severely. We propose and evaluate a curve-clustered sandwich covariance with a leverage adjustment for the penalized hat matrix with Satterthwaite degrees of freedom which brings coverage close to nominal in most settings. Its intervals for function-on-function effects are wide because REML's estimate is noisy. The same construction at the NCV estimate is narrow but undercovers. We therefore recommend fitting twice: NCV for the shape of the surface, REML with these intervals for inference. For estimated means, intercept functions and scalar covariate effects, one REML fit with these intervals usually suffices.

1 Introduction

Function-on-function regression relates a functional response $Y_i(t)$ to functional and scalar covariates (Ramsay and Silverman 2005; Müller and Yao 2008; J. S. Morris 2015). Its terms can include, among others, a *functional intercept* $\alpha(t)$, the linear predictor at zero covariates; *coefficient surfaces* $\beta(s,t)$ for functional covariates, which weight the covariate's value at position s in its effect on the response at t; and *coefficient functions* $\gamma(t)$ for t-varying effects of scalar covariates. In the penalized framework of Ivanescu et al. (2015) and Greven and Scheipl (2017), implemented in `pffr()` of the R package refund (Scheipl, Staicu, and Greven 2015; Goldsmith et al. 2024), each term is a penalized (tensor product) spline. The model is

*All code and text were generated by large language models in collaboration with the author; see the statement on the use of generative AI at the end of the main text.

fitted as one generalized additive model (Wood 2017) to the $N = GD$ stacked grid values of G curves with D points each, and the fit treats these N values as independent given the additive predictor.

Real world residual curves typically violate this assumption: they tend to be somewhat smooth deviations from the mean, or artifacts of misaligned features or slow drift, so values at grid points of each curve are correlated. This dependence has two consequences. First, smoothing-parameter selection by restricted maximum likelihood (REML), like other data-driven selection under correlated errors (Opsomer, Wang, and Yang 2001), undersmooths: the estimates, in particular of the coefficient surface, are more variable than necessary. Second, the model-based standard errors count N independent observations although the data carry much less information. Estimated intervals are then too narrow, and finer t-grids make them narrower without adding information.

This paper provides guidance on what users of `pffr()` (or other functional regression software) should report, and lays out the rationale for the defaults we have implemented. Analysts who fit such a model require estimates that show the shape of each effect faithfully and intervals whose error rates are close to nominal even under strong dependence or variance heterogeneity. We evaluate the available fits and covariances against both requirements, for Gaussian, count and binary responses, in a simulation with 152 settings and in a plasmode study that attaches the real residual curves of nine datasets to known "truths". Throughout, the model for the mean is taken to be correct and only the dependence is misspecified; what happens when the basis cannot represent the true effects is a separate question, which we touch on only briefly (Section 7).

Curve-blocked neighborhood cross-validation (NCV) (Wood 2024) chooses the smoothing parameters by leaving out one whole curve at a time. In the simulation's main settings, REML's relative error for $\beta(s,t)$ under dependence is two to four times that of this NCV fit. When the true $\beta(s,t)$ is zero and the errors are smooth, REML's estimate shows spurious structure, and NCV's stays much closer to zero. On the plasmode data, the error of either fit for $\beta(s,t)$ is large relative to the effect, possibly through weak identification: in directions that the functional covariate does not vary in, $\beta(s,t)$ is free to vary with hardly any effect on the fitted values (Scheipl and Greven 2016). NCV estimates the surface better for most "truths" on most datasets, but not always.

Under dependence, model-based intervals are far too narrow for fitted means and model terms, both in the simulation and on real residual curves, and more so on denser grids. A curve-clustered sandwich covariance with a leverage adjustment for the penalized hat matrix (CL2), computed after the usual REML fit (REML + CL2) and used with Satterthwaite degrees of freedom, brings coverage of fitted means and $\beta(s,t)$ close to nominal in the simulation's main settings, also with 20 curves, and on eight of the nine plasmode datasets; it falls short on the ninth.

Neither the REML nor the NCV fit gave both a good estimate and calibrated intervals for the surface. REML + CL2 intervals are wide because REML's estimate is noisy. Intervals centered at the NCV estimate are narrow but undercover, in the simulation mainly through smoothing bias, and the corrections we tried did not calibrate them on the plasmode data. These results show the classical tension between estimation and pointwise inference in nonparametric regression. An undersmoothed estimate has small bias relative to its standard error, so its intervals can hold their level; an estimate smoothed for accuracy needs an explicit bias correction, whose own variability then enters the interval (Hall 1992; Calonico, Cattaneo, and Farrell 2018). Here REML undersmooths through the dependence rather than by design, and in the simulation its intervals with CL2 cover. The tension is strongest for effects that vary along t and a second argument. For the functional intercept, the coefficient functions of scalar covariates and fitted means, REML + CL2 gives both a good estimate and intervals close to nominal, except for the intercepts and coefficient functions of binary responses and for counts under misregistration.

Our recommendation therefore depends on the goal, which in practice may mean fitting the model twice (Section 6). To learn the shape of a $\beta(s,t)$ surface, use curve-blocked NCV, our default, which typically smooths more than REML. For inference about that surface, use REML + CL2 with Satterthwaite degrees of freedom.

We make four contributions. (i) We give evidence on how much curve-blocked NCV improves the estimate of the coefficient surface over REML, and where it does not. (ii) We compute the CR2 leverage adjustment of Bell and McCaffrey (2002) with the non-idempotent hat matrix of penalized fits, combine it with the model-based smoothing-bias allowance and Satterthwaite degrees of freedom, and implement it in the refund R package. (iii) Our evaluation reports estimation error, width, detection of non-zero effects and false detections of a zero effect next to coverage, in synthetic and real-data-based settings of varying sizes and noisiness. (iv) We report negative results on intervals centered at the NCV estimate and on other attempted remedies in the supplement.

Related work. Within-curve dependence can be modeled directly: by functional mixed models with structured residual covariance (J. S. Morris and Carroll 2006; Meyer et al. 2015), by generalized least squares with an estimated residual covariance (Reiss, Huang, and Mennes 2010) and by curve-level functional random effects (Scheipl, Gertheiss, and Greven 2016); the last two underlie two of our comparators. Signal compression (Luo and Qi 2017) is a different estimator of the coefficient surface, and Cao, Wang, and Wang (2020) derive asymptotic normality for tensor-product spline estimators of coefficient surfaces. Correlated errors distort data-driven selection in scalar smoothing (Opsomer, Wang, and Yang 2001; Caron Parte, Dedecker, and Michel 2026); likelihood-based selection is less sensitive than generalized cross-validation (Krivobokova and Kauermann 2007). Leaving out whole curves avoids modeling the dependence (Rice and Silverman 1991), and neighborhood cross-validation (Wood 2024) makes this cheap.

For inference, pointwise and simultaneous bands for functional means and for function-on-scalar and scalar-on-function models are well developed (Degras 2011; Goldsmith, Greven, and Crainiceanu 2013; Choi and Reimherr 2018; Liebl and Reimherr 2023). Fast univariate inference (Cui et al. 2022) and penalized functional generalized estimating equations (Loewinger et al. 2026) address longitudinal functional data with many subjects. Park et al. (2018) estimate fixed effects under working independence and obtain their inference from a bootstrap of subjects. Working-independence estimators for clustered data are consistent but can be inefficient (Lin and Carroll 2000; Welsh, Lin, and Carroll 2002). Cluster-robust covariances need only independence across clusters (Liang and Zeger 1986; Cameron and Miller 2015; Zeileis, Köll, and Graham 2020; Hansen and Lee 2019). With few clusters, the leverage adjustment of Bell and McCaffrey (2002) and its extensions (McCaffrey and Bell 2006; Pustejovsky and Tipton 2018) reduce the downward bias of the sandwich (Kauermann and Carroll 2001). Here the number of coefficients (about 100) is comparable to the number of curves, so per-curve leverage is not negligible (MacKinnon, Nielsen, and Webb 2023; Carter, Schnepel, and Steigerwald 2017). Korolev (2026) adjusts the cluster-robust covariance for the penalty in semiparametric panel models fitted with mgcv.

2 Model, fits and covariances

2.1 Model and penalized fit

We work within the general functional regression framework of Greven and Scheipl (2017), which relates a feature of the conditional distribution of a functional response $Y_i(t)$ given covariate set $\mathcal{X}_i$ to an additive

predictor,

$$\xi(Y_i(t) \mid \mathcal{X}_i) = h(\mathcal{X}_i) = \sum_{\ell=1}^{L} h_\ell(\mathcal{X}_i). \tag{1}$$

The response is a function $Y_i(t)$ over an interval $\mathcal{T}$, with scalar responses as the special case of a single-point $\mathcal{T}$, and $\mathcal{X}_i$ can contain scalar and square-integrable functional covariates as well as grouping factors. The transformation ξ determines the modeled feature: for generalized functional additive mixed models (GFAMMs) (Scheipl, Gertheiss, and Greven 2016), the pointwise conditional mean under a link function, $\xi = g \circ \mathrm{E}$. Each partial predictor $h_\ell(\mathcal{X}_i)$ is a function over $\mathcal{T}$ with values $h_\ell(\mathcal{X}_i)(t)$ and depends on a subset of the covariates. Options include functional intercepts, linear or smooth effects of scalar and functional covariates, their interactions, and functional random effects, in which case the model also conditions on these latent processes.

In the following, we use a GFAMM with three partial predictors and no functional random effects. For curves $i = 1, \ldots, G$ observed at $t_1, \ldots, t_D$, the model is

$$g\{\mathrm{E}[Y_i(t) \mid X_i, v_i]\} = \eta_i(t) = \alpha(t) + \int_{\mathcal{S}} X_i(s)\, \beta(s,t)\, ds + v_i\, \gamma(t), \tag{2}$$

with link g, functional intercept $\alpha(t)$, coefficient surface $\beta(s,t)$ of the functional covariate $X_i(s)$ and coefficient function $\gamma(t)$ of the scalar covariate v_i. A smooth effect $f(x_i, t)$ of a scalar covariate can replace the functional term. The responses follow an exponential family, or an "extended" family such as the scaled t-distribution (Wood, Pya, and Säfken 2016), with mean $g^{-1}\{\eta_i(t)\}$ and variance $\phi V(\mu)$.

Each term is expanded in a penalized spline basis, with tensor product splines for multivariate terms. The stacked predictor is $\eta = \mathbf{X}\theta$ with penalty $\sum_r \lambda_r \theta^\top \mathbf{S}_r \theta$. Penalized iteratively reweighted least squares gives $\hat{\theta}$ for given smoothing parameters λ. Let $\mathbf{W}$ be the diagonal matrix of working weights $(d\mu/d\eta)^2/\{\phi V(\mu)\}$ at convergence and $\mathbf{S} = \sum_r (\lambda_r/\phi)\mathbf{S}_r$. The Bayesian and the frequentist covariance under the working model are

$$\mathbf{V}_p = (\mathbf{X}^\top \mathbf{W}\mathbf{X} + \mathbf{S})^{-1}, \qquad \mathbf{V}_e = \mathbf{V}_p \mathbf{X}^\top \mathbf{W}\mathbf{X}\, \mathbf{V}_p. \tag{3}$$

Their difference $\mathbf{V}_p - \mathbf{V}_e = \mathbf{V}_p \mathbf{S} \mathbf{V}_p$ is a prior-averaged allowance for smoothing bias (Wahba 1983; Nychka 1988; Marra and Wood 2012). For REML fits, mgcv adds a term for the uncertainty of $\hat{\lambda}$ and returns $\mathbf{V}_c$ (Wood, Pya, and Säfken 2016). The *model-based* intervals below use $\mathbf{V}_c$ for REML fits and $\mathbf{V}_p$ for NCV fits, because the $\mathbf{V}_c$ that mgcv returns for NCV fits in the version used here (an unpatched pre-release build of 1.9-5) is not a valid covariance (Supplement S6). The two model-based intervals therefore contain different uncertainty terms. All these covariances assume that all N values $Y_i(t_j)$ are conditionally independent given $\mathcal{X}_i$.

2.2 Smoothing-parameter selection

REML maximizes the Laplace-approximate restricted likelihood of the working model (Wood 2011). Estimating a term averages residual curves with weights that depend on the covariates. Within a curve, neighboring residuals are correlated, so pooling across grid points reduces the noise in this average much less than assumed by the working likelihood, which counts N independent values. We believe this excess noise, to the extent that the chosen spline basis can represent it, is picked up as signal by REML when it selects too little smoothing. REML's excess error grows under dependence and on denser grids (Section 4.1), which fits this reading.

Curve-blocked NCV. Neighborhood cross-validation (Wood 2024) minimizes

$$\mathscr{C}(\lambda) = \sum_{k=1}^{m} \sum_{(i,j)\in\mathcal{N}_k} \operatorname{dev}\big(y_{ij}, \hat{\theta}_\lambda^{-\mathcal{N}_k}\big), \tag{4}$$

the deviance of predictions for the points in $\mathcal{N}_k$ from the fit without them, with $y_{ij} = Y_i(t_j)$. We use one neighborhood per curve, $\mathcal{N}_k = \{(k, j) : j = 1, \dots, D\}$, so $m = G$ and each neighborhood has as many points as its curve. The criterion then measures how well a fit to the other curves predicts a new curve from its covariates (Rice and Silverman 1991). If residual curves are independent across curves, no information about the left-out curve is contained in the noise a fit follows in the other curves, so following its pattern worsens predictions and the criterion selects stronger smoothing even without an explicit model of the dependence. This holds only for the modes of variation of a term that directly affects predictions, i.e., not for changes such as $\beta(s,t) + \delta(s)f(t)$ with $\delta(s)$ in the kernel of the covariance operator of $X(s)$ (Scheipl and Greven 2016). Where curves are dependent on each other, the neighborhoods should be sub-groups that are independent of each other, such as all curves of a given subject. Spatial block neighborhoods are evaluated on the three plasmode datasets with dependent source curves (Section 5; Supplement S2.3); subject neighborhoods were not evaluated in this work.

No curve is actually refitted: $\hat{\theta}^{-\mathcal{N}_k}$ is approximated by a single Newton step based on the full fit, which is exact only for Gaussian responses with identity link and given λ, and $\mathscr{C}$ is minimized over λ with analytic derivatives (Wood 2024). Gaussian fits with $G = 100$ and $D = 61$ using curve-blocked NCV took a median 8.6 s against 2.0 s for REML.

2.3 Curve-clustered covariances

The curve-clustered sandwich replaces the variance part of $\mathbf{V}_p$ by one that does not assume conditional independence within curves. It does assume conditional independence across curves as well as a correct model for the mean. Write $\tilde{\mathbf{X}}_i = \mathbf{W}_i^{1/2}\mathbf{X}_i$ for the weighted design rows of curve i, $\mathbf{z}_i$ for its Pearson residuals, and $\tilde{\mathbf{H}} = \tilde{\mathbf{X}}\mathbf{V}_p\tilde{\mathbf{X}}^\top$ for the penalized hat matrix. The CL2 covariance is

$$\hat{\mathbf{V}}_{\mathrm{CL2}} = \frac{G}{G-1}\,\mathbf{V}_p\Big(\sum_{i=1}^{G}\mathbf{U}_i\mathbf{U}_i^\top\Big)\mathbf{V}_p + (\mathbf{V}_p - \mathbf{V}_e), \qquad \mathbf{U}_i = \tilde{\mathbf{X}}_i^\top\mathbf{R}_i^{-1/2}\mathbf{z}_i, \tag{5}$$

with $\mathbf{R}_i = [(\mathbf{I} - \tilde{\mathbf{H}})(\mathbf{I} - \tilde{\mathbf{H}})^\top]_{ii}$, the block of curve i in the covariance of the residuals under the working model. CR1, the usual cluster-robust covariance (Liang and Zeger 1986), sets $\mathbf{R}_i = \mathbf{I}$. The adjustment $\mathbf{R}_i^{-1/2}$ is the bias-reduced linearization of Bell and McCaffrey (2002) (CR2), with the penalized hat in place of a projection. Because $\tilde{\mathbf{H}}$ is not idempotent, $\mathbf{R}_i \neq \mathbf{I} - \tilde{\mathbf{H}}_{ii}$; the full block is computed. We keep the factor $G/(G-1)$, floor the eigenvalues of $\mathbf{R}_i$ at 10^{-6}, and write CL2 to mark these departures from CR2. The second term of Equation 5 keeps the model-based allowance for smoothing bias, so CL2 replaces only the variance part of $\mathbf{V}_p$. The variance part estimates the sampling variance of $\hat{\theta}$ around its penalized target; the added $\mathbf{V}_p - \mathbf{V}_e$ is a prior-averaged allowance for the distance between that target and the truth which does not provide guaranteed coverage of every fixed truth (Appendix A.3).

The sandwich must use the penalized bread $\mathbf{V}_p$ and the penalized hat. General-purpose cluster-robust software does not provide this combination. The clustered covariance of the sandwich package (Zeileis et al. 2024) uses the penalized bread of an mgcv fit and reproduces CR1 without the allowance, but its leverage adjustments are restricted to unpenalized (generalized) linear models, for whose projection hat

CR2 is also defined in clubSandwich (Pustejovsky 2024). Limited asymptotic consistency and validity results for fixed smoothing parameters are provided in Appendix A.5, but in the realistic regimes we study here (about as many spline coefficients as curves, non-negligible penalties and estimated smoothing parameters), the required assumptions do not apply. CL2 is a finite-sample procedure whose behavior is established empirically below.

Satterthwaite degrees of freedom. For any contrast $\mathbf{a}^\top\theta$, the variance part of Equation 5 is a sum of squares $\sum_i(\mathbf{q}_i^\top \mathbf{z}_i)^2$ with $\mathbf{q}_i = \mathbf{R}_i^{-1/2}\tilde{\mathbf{X}}_i\mathbf{V}_p\mathbf{a}$, up to the factor $G/(G-1)$. Under a Gaussian working model with fixed λ and no smoothing bias, the residuals are $(\mathbf{I}-\tilde{\mathbf{H}})\varepsilon$ with independent standard normal ε, and matching the first two moments of the quadratic form to a scaled χ^2_ν gives $\nu = (\operatorname{tr}\Gamma)^2/\operatorname{tr}(\Gamma^2)$ with $\Gamma_{ih} = \mathbf{q}_i^\top[(\mathbf{I}-\tilde{\mathbf{H}})(\mathbf{I}-\tilde{\mathbf{H}})^\top]_{ih}\mathbf{q}_h$ (see Appendix A.7 and Bell and McCaffrey 2002; Pustejovsky and Tipton 2018). The interval uses the t_ν quantile at each grid point. We use the same ν for all families (Appendix A.7 discusses this choice). The degrees of freedom ignore non-normality of the responses, the bias allowance, the dependence and the selection of λ; they are a reference distribution, not an exact pivot.

The same constructions apply to both fits: *REML + CL2* is Equation 5 at the REML estimate and *NCV + CL2* at the NCV estimate.

2.4 Estimands and performance measures

Four estimands are evaluated on fixed grids: the conditional mean $\mathrm{E}(Y \mid X)$ for a fixed set of 50 subjects' covariates, on the response scale; $\beta(s,t)$ (or $f(x,t)$); $\alpha(t)$; and $\gamma(t)$, on the link scale. Pointwise intervals are estimate ± critical value × standard error, transformed by g^{-1} for the mean. Every method is judged on accuracy of the estimate, calibration and informativeness (T. P. Morris, White, and Crowther 2019):

- *Coverage*: the share of replicates whose interval contains the truth at a grid point, averaged over the J evaluation points of the estimand. Its Monte Carlo standard error is the standard deviation over replicates of the grid-averaged coverage divided by the square root of the number of replicates. The *lower tail* is the 5% quantile of pointwise coverage over the grid. Each pointwise coverage is itself estimated from 200 replicates, so the lower tail lies below the nominal level even for a perfectly even interval: with coverage 0.95 at every point, it would be about 0.925.
- *Relative error*: $\{\sum_j(\hat{\vartheta}_j - \vartheta_j)^2/\sum_j \tilde{\vartheta}_j^2\}^{1/2}$ over grid points j, for estimate $\hat{\vartheta}$ and truth ϑ, with $\tilde{\vartheta} = \vartheta$ for β and γ and $\tilde{\vartheta}$ the truth centered at its grid mean for the mean and $\alpha(t)$, whose level is not the effect. A value of 1 means the error is as large as the effect. We report its median over replicates.
- *Relative half-width*: the root-mean-square half-width of the intervals over the grid, divided by $(\sum_j \tilde{\vartheta}_j^2/J)^{1/2}$, the root-mean-square size of the (centered) truth.
- *Detection rate*: among grid points where the absolute true value exceeds 25% of its maximum, $|\vartheta_j| > 0.25\max_k|\vartheta_k|$, the share whose interval excludes zero on the side of the truth.
- *Interval score* (Gneiting and Raftery 2007): for an interval $[l,u]$ and true value ϑ, $(u-l)+(2/0.05)\{(l-\vartheta)_+ + (\vartheta-u)_+\}$, averaged over the grid and the replicates; lower is better. It is a proper scoring rule for predictive quantile intervals; against fixed truths over repeated datasets it rewards an accurate center and a narrow width together and is not a calibration criterion. We report it relative to REML + CL2 and interpret it in context of the other measures.

The supplement adds wrong-sign rates and studentized errors. Comparisons of estimators use the ratio of the two fits' median relative errors; where ratios of mean squared errors are given, they are labeled as such. In Section 4, both are REML to NCV. Table 1 lists the principal interval constructions and where each was evaluated.

Table 1: Principal interval constructions. Covariance, bias-allowance and critical-value variants are specified in Appendix A.6; comparator and NCV-centered constructions in Supplement S3 and S4. All CL2 variants cluster by curve, use the penalized bread $\mathbf{V}_p$ and include the allowance $\mathbf{V}_p - \mathbf{V}_e$ unless called frequentist. Full block: $\mathbf{R}_i$ of Equation 5; shortcut: $\mathbf{R}_i \approx \mathbf{I} - \tilde{\mathbf{H}}_{ii}$.

interval	fit	covariance	critical value	evaluated in
model-based	REML	mgcv's $\mathbf{V}_c$	z	all settings, plasmode
model-based	NCV	$\mathbf{V}_p$	z	all settings (supplement)
REML + CL2	REML	CL2, full block, + $(\mathbf{V}_p - \mathbf{V}_e)$	z	all settings, plasmode
REML + CL2, Satterthwaite	REML	as REML + CL2	t_ν (Satterthwaite)	main factorial (default signal), extensions, plasmode
NCV + CL2	NCV	CL2, full block, + $(\mathbf{V}_p - \mathbf{V}_e)$	z	all settings, plasmode
CR1, CL2 shortcut, frequentist CL2, t_{G-1}	REML	variants of CL2	z or t_{G-1}	main factorial, default signal

3 Simulation design

The simulation follows the model in Equation 2 with a functional covariate whose spectrum matches the corpus-callosum profiles of the DTI data in refund (152 settings, 200 replicates each, 30,400 fits per selector; Appendix B.1 lists all blocks of settings). The truths are fixed smooth functions, projected onto the used spline bases so that they are representable without error; "rough" variants add higher-frequency components. Responses are Gaussian (signal-to-noise $R^2 = 0.2, 0.5, 0.8$), Poisson and Bernoulli at each grid point (two signal-to-noise levels on the link scale), with scaled t, Beta and Negative-Binomial responses as extensions.

Within-curve errors are zero-mean processes with correlation kernel R, used directly for Gaussian responses and as the latent layer of a Gaussian copula otherwise, so that every marginal remains in the fitted family. So for count and binary data, the dependence is a copula construction, and the support for CL2 with these families is empirical within that construction. Three error processes form the main factorial:

- *independent*: $R(t, t') = \mathbb{1}\{t = t'\}$;
- *rough stationary*: $R(t, t') = 0.8 \exp(-|t - t'|/0.089) + 0.2\, \mathbb{1}\{t = t'\}$;
- *smooth non-stationary*: the residual correlation of the DTI regression in refund (Figure 13 in Appendix B.1).

The main factorial crosses these processes with the family, $G \in \{40, 100\}$ and the signal level at $D = 61$ grid points. Each extension changes one aspect of this design (Appendix B.1 lists the settings of every block): the main factorial with $G = 20$, a four times denser grid, a rough truth, a coefficient surface that is zero everywhere ($\beta \equiv 0$; Gaussian and binary responses, independent and smooth errors), a smooth effect $f(x, t)$ in place of the functional term $\int X(s)\beta(s, t)ds$, sign-changing and heteroskedastic errors, misregistration (each curve observed on its own clock / deformed elastically), a low-rank functional covariate and larger bases.

The fitted bases are cubic B-splines with first-order difference penalties (mgcv's `ps` basis with `m = c(2, 1)`), so the penalty null space contains the constants along each direction: $k = 12$ along t for $\alpha(t)$ and $\gamma(t)$ and an 8×10 tensor product for $\beta(s, t)$, about 100 coefficients in total; the larger bases halve and quarter the knot spacing. Results with second-order penalties were not computed.

Each replicate is fitted by REML and by curve-blocked NCV; none of the fits failed. For NCV fits of scaled-t responses, mgcv often reported that the search over the smoothing parameters ended without meeting its convergence test (88% under smooth errors, 68% under independent errors, but results were mostly stable under refits from different starting values indicating flat optima), and rarely in other families; REML fits

were not affected (settings with at least 40 curves and a non-zero surface; Appendix B.1). Random draws depend only on the replicate, so contrasts between methods and settings are paired. The Monte Carlo standard error of a grid-averaged coverage is 0.21–0.33 pp (percentage points) for REML + CL2 in the dependent main-factorial settings with 100 curves.

4 Simulation results

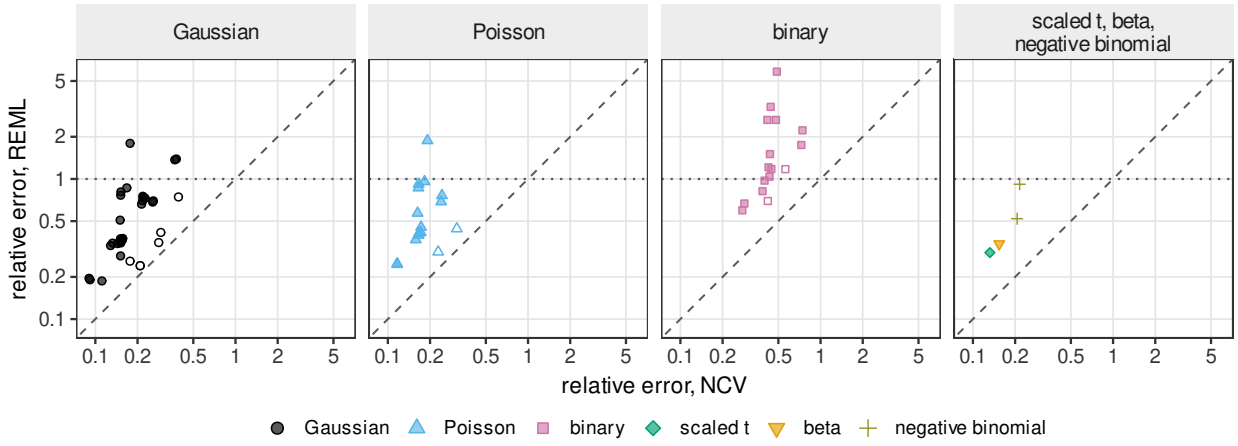


Figure 1: Relative error of the REML and the NCV estimate of $\beta(s,t)$ in all simulation settings with dependent errors, 40 or 100 curves and a non-zero surface (one symbol per setting; panels by family on common axes; median over replicates; log scales; open symbols: rough truth; dashed: equal error; dotted: error as large as the effect).

4.1 Estimation: REML versus NCV

Under independent errors, both selectors estimate $\beta(s,t)$ well: the median relative error over the settings is 0.16 for REML and 0.12 for NCV. Under dependence, REML's error rises to 0.67 (maximum 5.8) and NCV's to 0.21 (Figure 1). In the main factorial, REML's median relative error is 2.1–3.7 times NCV's under dependence, at every signal level (Table 2). The ratio is larger on a denser grid and with larger bases, and smaller with a rough truth and for a smooth effect $f(x,t)$ in place of the functional term. NCV estimates $\beta(s,t)$ better also under independent errors when the truth is smooth (ratio of mean squared errors 1.1–1.8 for Gaussian and Poisson responses; 0.87–1.02 for binary responses), but not when it is rough (0.85–0.98 with 100 curves). Part of the gap therefore reflects the smooth truths rather than the dependence: REML selects the smoothing parameters under the prior implied by the penalty, and undersmoothing a truth that is smoother than this prior is a known property of likelihood-based selection (Wahba 1985). Leaving out single grid points instead of whole curves does not give NCV's gain (Supplement S1.5).

REML's surface uses 1.9–2.3 times NCV's effective degrees of freedom (EDF) under dependence (1.1–1.3 under independence). Its excess error is variance, not bias: squared bias is 0.01 of REML's mean squared error for β and 0.12–0.20 of NCV's (100 curves, smooth truth). Larger bases make it worse: with the largest basis, REML's surface EDF under smooth errors grows from 27–42 to 113–176 across families, NCV's from 12–21 to 18–41 (Supplement S5).

Table 2: Ratio of the median relative errors of the REML and the NCV estimate of $\beta(s,t)$ (REML to NCV; smooth effect: $f(x,t)$): range over the settings of each block. The extensions have dependent errors unless stated; misregistered curves have independent errors.

settings	REML / NCV
main factorial, dependent errors	2.1–3.7
main factorial, independent errors	1.1–1.4
20 curves, dependent errors	2.5–4.7
20 curves, independent errors	1.0–1.3
four times denser grid	5.0–7.4
larger bases	5.2–12.0
rough truth	1.3–2.1
misregistration	1.1–1.9
smooth effect in place of the functional term	1.3–1.7

Estimates for the mean, the intercept and the scalar effect depend much less strongly on the selector. Under independent errors, the median relative error over the settings is 0.11 (REML) and 0.10 (NCV) for the mean, 0.16 and 0.16 for $\alpha(t)$, and 0.11 and 0.12 for $\gamma(t)$. Under dependence, it is 0.28 and 0.23, 0.31 and 0.28, and 0.22 and 0.22 (Figure 3a). In the dependent main-factorial settings, the ratio of REML's to NCV's median relative error is 1.1–1.3 for the mean, 0.94–1.30 for $\alpha(t)$ and 0.98–1.13 for $\gamma(t)$, against 2.1–3.7 for β. NCV thus gains modestly for the mean (ratio of mean squared errors 1.2–1.7 with 100 curves) and little for the other two terms; for $\alpha(t)$ the ratio stays at or below 1.09 outside binary responses. REML also has the heavier tail: 4.3% of replicates have a relative error above 1 for the mean, against 0.7% for NCV, with smaller differences for $\alpha(t)$ (10.8% and 8.5%) and $\gamma(t)$ (3.8% and 3.2%).

Stability of the smoothing parameters. GCV can have multiple minima and more variable smoothing parameters than REML (Reiss and Ogden 2009), and prediction-error criteria occasionally undersmooth severely (Wood 2011). Curve-blocked NCV, which these results do not cover, shows a similar tail. With 40 or 100 curves (Appendix B.1), the first of the surface's two smoothing parameters fell more than a factor of $e^3 \approx 20$ below its setting's median in 2.2% of NCV fits and in no REML fit under independent errors, and REML estimated the surface better on 99% of those replicates. Such fits made up 6.0% of NCV fits under dependence, apart from misregistration, but there NCV still estimated the surface better on 97% of those replicates. They have 2.6 times their setting's median relative error and account for 31% of NCV's squared error, but they do not explain its undercoverage: leaving them out moves NCV + CL2 only from 0.900 to 0.904. Convergence flags do not detect them: outside scaled-t responses, only 1 of 542 was flagged.

Neither fit is uniformly the more stable under dependence: the median standard deviations of the two log smoothing parameters over replicates are 1.34 and 0.83 for NCV and 2.09 and 0.37 for REML (under independence, 0.91 and 0.75 against 0.14 and 0.10). Restarts of flagged scaled-t fits can sometimes result in materially different fits (Appendix B.1); multiple starts were not evaluated as a fitting procedure.

Alternative models. Explicit models for residual dependence also reduce REML's surface estimation error. Two such methods were compared with NCV on the same data, in 4 Gaussian main-factorial settings with dependent errors as well as 12 plasmode combinations (Supplement S3.2): a functional random effect for each curve based on the eigenfunctions of the REML-estimated residual curves (`pcre()` in refund), and generalized least squares (GLS) with a residual covariance estimated from the REML residual curves by functional principal components (FPCA) plus white noise (GLS-FPCA). Their mean squared error for $\beta(s,t)$, relative to NCV's (geometric mean over settings), is 1.75 (pcre) and 1.79 (GLS) in the simulation, where REML's is 7.4, and 0.72 and 0.68 on the plasmode data (midpoint truth, defined in Section 5, with REML residuals; the GLS covariance is estimated from the residual curves of a REML pilot fit to each simulated dataset, and because these errors are sign-flipped REML residuals of the original fit, those pilot residuals

closely resemble the curves that generate them). Both are thus much better than REML, worse than NCV in the simulation and comparable to NCV in that plasmode comparison. Including the REML pilot fit, a pcre fit took 2.3–20 times as long as an NCV fit in the simulation and 0.9–20 times as long on the plasmode data; a GLS fit took 0.6–0.7 and 0.4–0.9 times as long (medians per setting). The GLS algorithm – only applicable for Gaussian data – is given in Supplement S3.2.

4.2 Intervals for the REML fit

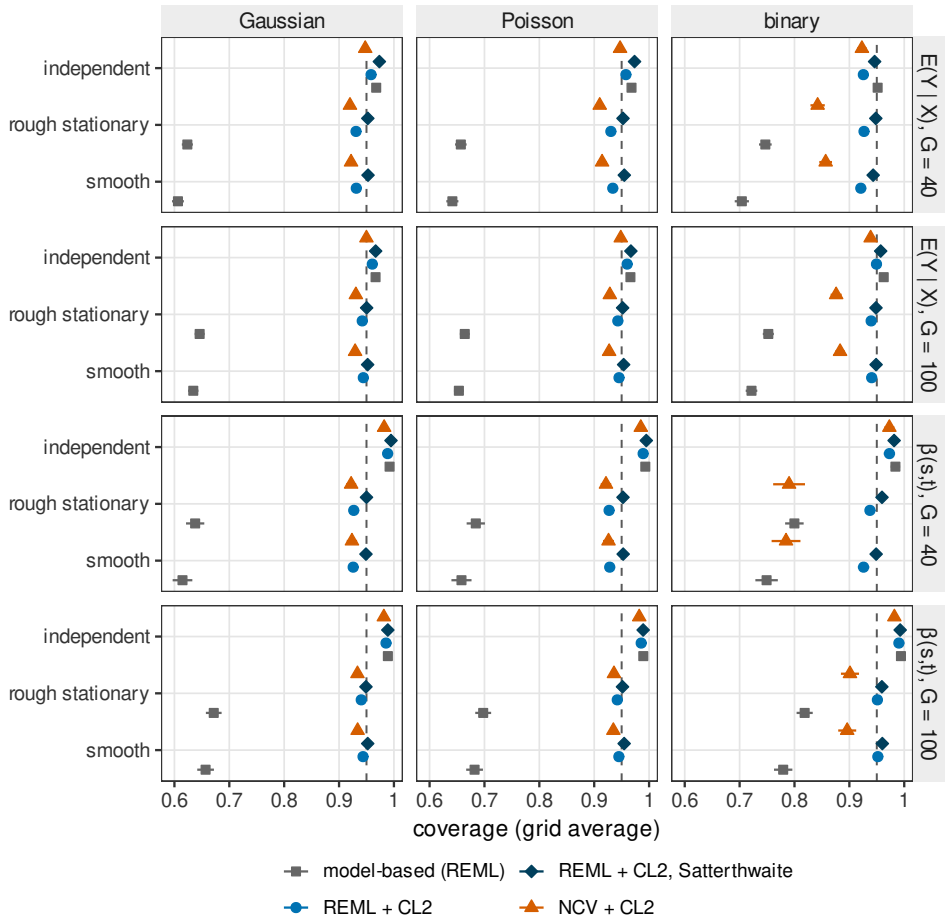
Model-based intervals undercover in every dependent setting and every family (Figure 2a): 0.63–0.82 for the mean and $\beta(s,t)$ with 100 curves. On a four times denser grid ($D = 241$), their coverage of β drops to 0.35–0.45. REML + CL2 covers the mean and β at 0.94–0.95 with 100 curves and stays at 0.94 for β at $D = 241$. The same holds for $\alpha(t)$ and $\gamma(t)$ of Gaussian and count responses, for scaled t, beta and negative-binomial responses, for the smooth-effect model, for sign-changing and heteroskedastic errors, and for misregistered Gaussian curves (Supplement S1.1–S1.3; misregistered Gaussian curves: mean and β 0.93–0.95). In the dependent main-factorial settings, REML + CL2's coverage is also much more even over the grid: the lower tail for the mean and β is 0.91–0.93 with 100 curves and 0.89–0.91 with 40, against 0.54–0.76 for the model-based intervals with 100 curves.

Under independent errors with 100 curves, CL2 gives up 0.3–1.1 pp of coverage against the model-based intervals for the mean and β (averaged over signal levels within each family). For binary responses with 40 curves at the default signal, it gives up 2.6 pp for the mean, where the model-based intervals do not overcover (REML + CL2 0.93, model-based 0.95), and 4.4 pp and 4.1 pp for $\alpha(t)$ and $\gamma(t)$ (Appendix A.6). For $\beta(s,t)$, the intervals overcover under independent errors (Figure 2a), because the allowance exceeds the smoothing bias of a smooth truth (Appendix A.3): REML + CL2 with the Satterthwaite critical value covers 0.98–0.99 (Appendix A.7).

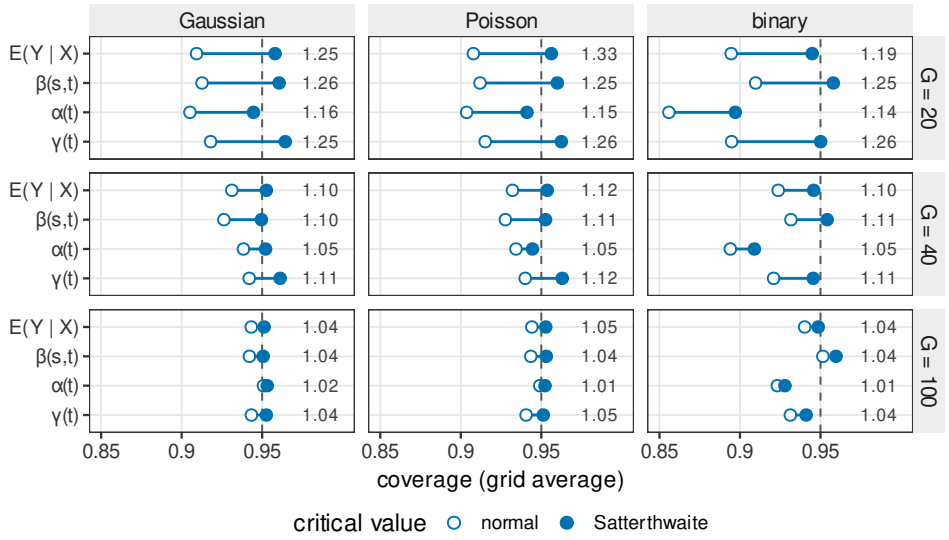
Fewer curves. With 40 curves, REML + CL2 covers the mean and β at 0.92–0.94. With 20 curves, it covers the mean, β and γ at 0.88–0.93 and $\alpha(t)$ at 0.83–0.91. Satterthwaite degrees of freedom recover most of this shortfall (Figure 2b): with 20 curves and the default signal, coverage of the mean, β and γ rises from 0.89–0.92 to 0.94–0.96, at up to 1.33 times the width; with 100 curves, the change is small. The binary intercept stays short (0.90 with 20 curves, 0.91 with 40). The degrees of freedom are about a third of $G - 1$ for the mean, β and γ, and larger for $\alpha(t)$; Appendix A.7 explains why. In the extensions, they never lower coverage. There, with Satterthwaite degrees of freedom, coverage stays below 0.93 in 13 of 291 combinations of setting and estimand, all under misregistration or for binary intercepts and scalar effects (Supplement S1.7).

CL2 and CR1. Under dependence and with the normal critical value, CL2 covers the mean and β 3.1–4.2 pp more than CR1 with the same allowance with 40 curves, at 1.1–1.2 times its width, and 1.0–1.4 pp more with 100, at 1.04–1.06 times (Appendix A.6). For fixed smoothing parameters and a fixed basis, the two covariances agree to first order as the number of curves grows (Appendix A.5). CR1 was not run with 20 or with more than 100 curves. The leverage adjustment costs a median of 1.0 s per fit with 100 curves.

Where CL2 falls short. For binary responses, REML + CL2 covers the functional intercept and the scalar effect at 0.88–0.94 under dependence and 0.86–0.95 under independent errors. Under independent errors with 40 curves, the intercept's studentized error is centered but too spread out, which is compatible with a standard error that is too small, although local biases of opposite sign are not excluded. Under dependence, it is also shifted (Supplement S1.4). Fixing the smoothing parameters at their typical value raises coverage of the binary $\gamma(t)$ under independent errors from 0.89 to 0.93 and of $\alpha(t)$ under smooth errors from 0.88 to 0.92 (40 curves), so part of the shortfall is likely due to uncounted smoothing-parameter variability; $\alpha(t)$ under independent errors stays at 0.87 (Appendix A.6, Supplement S1.4). With 20 curves and independent



(a) Coverage of the mean and of $\beta(s,t)$



(b) Normal and Satterthwaite critical values for REML + CL2

Figure 2: Grid-averaged coverage of nominal 95% pointwise intervals, main factorial at the default signal level (dashed: 0.95). (a) By family (columns), error process, estimand and number of curves G (rows). REML + CL2 and NCV + CL2 use the normal critical value; REML + CL2, Satterthwaite, is the recommended interval. Bars: ±2 Monte Carlo SE. NCV + CL2 is discussed in Section 4.3. Coverage for $\alpha(t)$ and $\gamma(t)$: Figure 11 in Appendix A.6; summary over all settings: Figure 3a. (b) Dependent errors (mean over the two processes). Numbers at the right: width of the Satterthwaite interval relative to the normal one.

errors, every interval undercovers binary $\alpha(t)$ and $\gamma(t)$, the model-based ones included (0.80–0.93), so this shortfall is not specific to the sandwich.

For misregistered counts, the working model for the mean is wrong: REML + CL2 covers the mean of misregistered Poisson responses at 0.90–0.91 (model-based 0.56–0.63).

4.3 The trade-off for the coefficient surface

In the simulation, REML's estimate of $\beta(s,t)$ is noisy but has little bias, and CL2 measures its spread, so its intervals cover. NCV's estimate has smaller error but more bias, beyond what the allowance $\mathbf{V}_p - \mathbf{V}_e$ covers, so its narrower intervals cover less. A more accurate estimate therefore need not give better coverage.

Under dependence, the median half-width of REML + CL2 for β is 1.4 times the size of the effect, and it detects only 0.54 of the grid points where the true value exceeds 25% of its maximum (Figure 3a, Figure 3b). Its intervals are 3.7 times as wide as those of NCV + CL2, which detects 0.95 but covers less (median 0.92, lower tail 0.84). These widths and detection rates are for the normal critical value. In the main factorial at the default signal, the Satterthwaite critical value widens REML + CL2 by a further factor of 1.04–1.11 (Figure 2b). Its detection rate for $\beta(s,t)$ under dependence then falls by at most 6 pp, to 0.11–0.49 with 40 curves and 0.25–0.78 with 100. The interval score of NCV + CL2 is 0.41 times REML + CL2's: the score rewards its accurate, narrow intervals more than it punishes the additional misses. We nevertheless do not recommend NCV + CL2 for inference, because its coverage is too low on the real residual curves (Section 5).

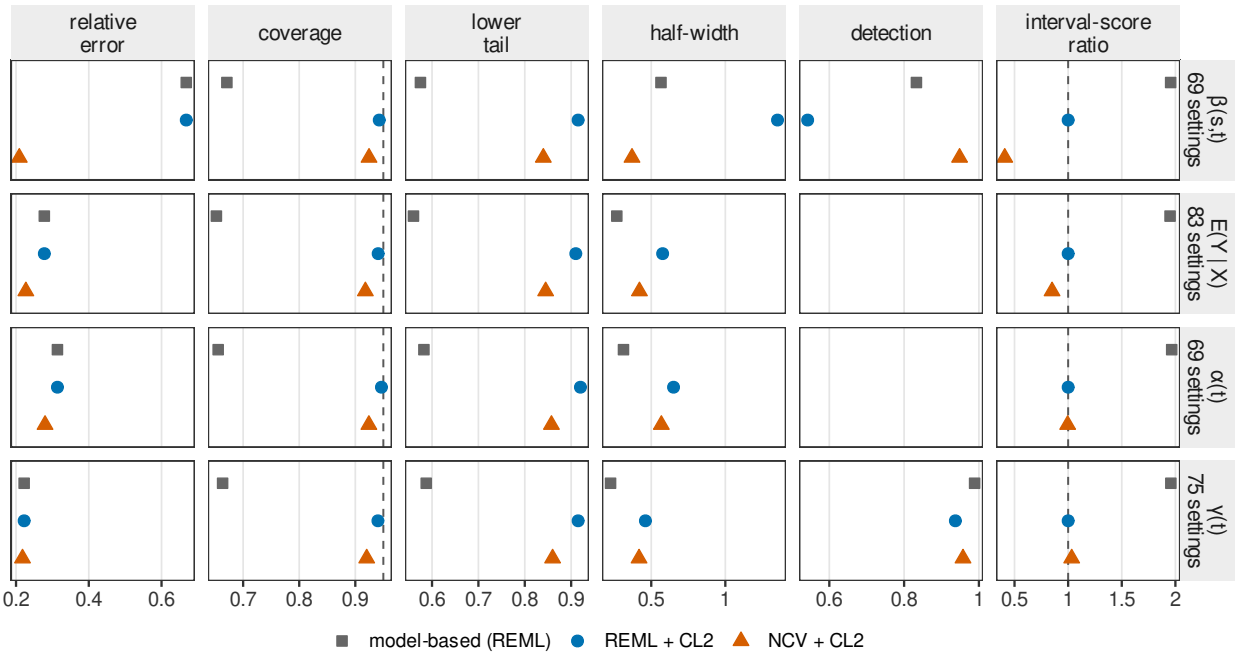
In the simulation, the shortfall of NCV + CL2 is mainly smoothing bias: subtracting the bias of the NCV estimate, estimated from the replicates, raises its coverage from 0.90–0.94 to 0.93–0.96 in the main factorial with 100 curves. NCV + CL2 covers less with 20 curves and dependent errors (0.71–0.92 for the four estimands), with a rough truth (0.80–0.90; REML + CL2: 0.94–0.95) and for binary responses with 40 curves (0.78–0.90).

The trade-off grows with the basis: with the largest basis and smooth errors, REML + CL2's half-width for β reaches 4–12 times the effect and its detection rate falls to 0.04–0.12, while its coverage stays at 0.95–0.96. Wrong-sign exclusions are rare for REML + CL2 (at most 2.3% of the grid in any dependent setting).

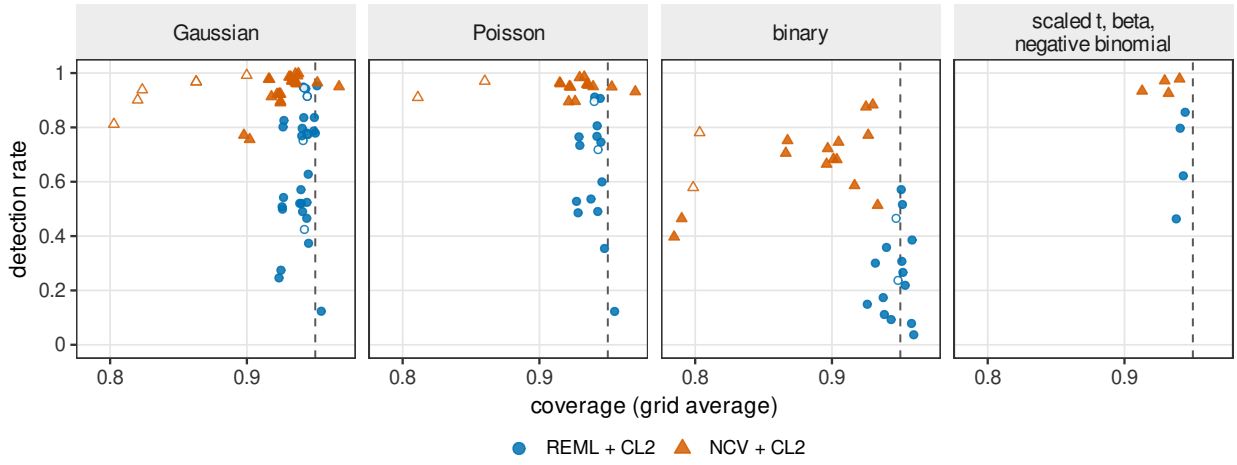
For the mean, $\alpha(t)$ and $\gamma(t)$, the trade-off is mild: REML + CL2's intervals are wider than NCV + CL2's (mean: 1.4 times), but they cover, the estimates of $\alpha(t)$ and $\gamma(t)$ are about as accurate, and $\gamma(t)$ is detected at 0.94.

The model-based intervals of pcre and GLS with an FPCA residual covariance show the trade-off from the other side. In the 4 Gaussian main-factorial settings with dependent errors, they cover β at 0.93–0.97, at 0.5–0.7 times REML + CL2's width and about half its interval score, but pcre's intervals cover the mean at only 0.88–0.91 and $\alpha(t)$ at 0.51–0.59. On real residual curves, both cover β at only 0.59–0.86 (Section 5).

CL2 also applies to these fits: GLS whitens the data within curves, so curves stay independent clusters, and pcre + CL2 is CL2 on the pcre fit (Supplement S3.2). Under dependent synthetic errors, GLS-FPCA + CL2 covers β at 0.96, at 0.6 times REML + CL2's width, and pcre + CL2 repairs pcre's intervals for $\alpha(t)$ (0.56 to 0.95). On the real residual curves, GLS-FPCA + CL2 covers β at 0.82 (0.65–0.90 by dataset) and pcre + CL2 at 0.83, against 0.93 for REML + CL2. GLS with the raw residual covariance fails with or without CL2 (0.65 under dependent synthetic errors). Both pcre and GLS-FPCA thus estimate better than REML (Section 4.1), and the sandwich repairs much, but not all, of their undercoverage.



(a) Summary per estimand and interval



(b) Coverage and detection rate for $\beta(s,t)$ per setting

Figure 3: All simulation settings with dependent errors, 40 or 100 curves and a non-zero surface (dashed: 0.95 for coverage, 1 for the ratio). (a) Median over settings of the relative error of the estimate, the grid-averaged coverage, the lower tail (5% quantile of pointwise coverage), the relative half-width and the detection rate, and the interval score relative to REML + CL2 (geometric mean over settings). The number of settings differs by estimand: $\beta(s,t)$ and $\alpha(t)$ are not scored in the smooth-effect model, and only the mean is scored for misregistered count and binary responses. Detection is defined for $\beta(s,t)$ and $\gamma(t)$ only. Values: Supplement S7. (b) One symbol per setting and interval; panels by family on common axes; open symbols: rough truth.

4.4 A zero coefficient surface

With $\beta \equiv 0$, the share of grid points at which an interval excludes zero is the pointwise false-positive rate, nominally 0.05 (Figure 4). Under smooth errors, it is 0.05–0.08 for REML + CL2 with the normal critical value and 0.05 with Satterthwaite degrees of freedom, against 0.11–0.15 for NCV + CL2 and 0.23–0.57 for the model-based intervals of either fit. Under independent errors, REML + CL2 has 0.04–0.06, except 0.095 for Gaussian responses with 40 curves (0.075 with the Satterthwaite critical value; NCV + CL2: 0.096). There, REML shrinks the surface to near the null space of its penalty in most replicates, and in some of these the interval excludes zero over much of the grid, a known weakness of intervals for terms shrunk to their null space (Marra and Wood 2012).

Under smooth errors, REML's estimate of the zero surface has 0.4–2.3 times the root-mean-square size of the non-zero surface in the corresponding setting: spurious structure. NCV's estimate is 5–7 times smaller. Under independent errors, both are close to zero.

Pointwise intervals do not answer whether $\beta(s,t)$ is zero everywhere: under smooth errors, 0.72–0.83 of the replicates have at least one grid point where the Satterthwaite interval excludes zero, and 0.33–0.41 have more than 5% of the grid excluded. That question needs simultaneous bands or global tests, which this study does not evaluate.

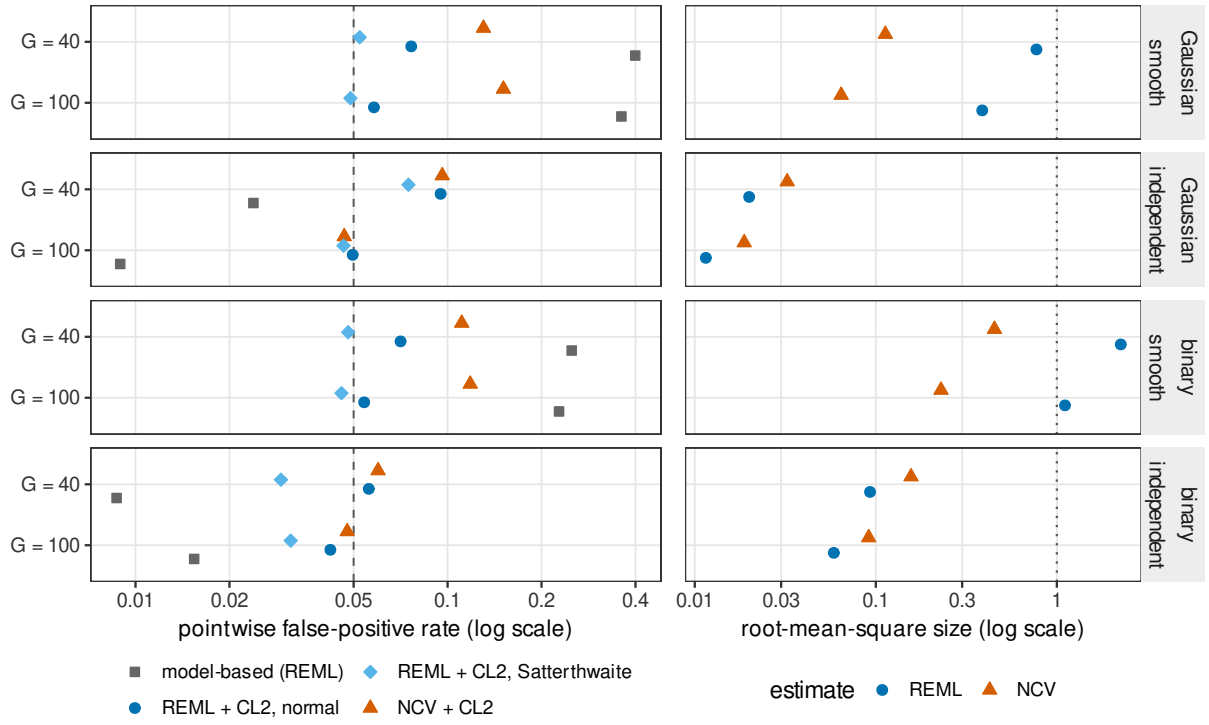


Figure 4: Zero surface ($\beta \equiv 0$), default signal, by family, errors and number of curves G (log scales). Left: pointwise false-positive rate of the model-based REML interval, REML + CL2 with the normal and the Satterthwaite critical value and NCV + CL2 (dashed: nominal 0.05). Right: root-mean-square size of the REML and NCV estimates relative to the non-zero surface of the corresponding setting (dotted: 1). Values: Supplement S1.8 and S7.

4.5 Other intervals and fits

We also evaluated the following constructions.

- *Intervals centered at the NCV estimate.* We tried widening by the NCV–REML difference, the NCV estimate with REML + CL2's half-width, a one-step bias correction of the NCV estimate and a cluster-robust term for the variability of the selected smoothing parameters, alone and combined. The combination is a frozen-map approximation: it propagates the smoothing-parameter term through the bias-correction map but ignores how that map changes with the smoothing parameters. In Gaussian settings, it covers $\beta(s,t)$ at 0.97 (smooth truths) and 0.94 (rough truths), but none of these constructions held its level across the plasmode datasets (Section 5.2; Supplement S4.1).
- *mgcv's neighborhood covariance for NCV fits.* For curve blocks, the unpatched mgcv build used here returns the Bayesian $\mathbf{V}_p$ or, with `jackknife = TRUE`, essentially $\mathbf{V}_p - \mathbf{V}_e$; it computes this covariance correctly only for per-point neighborhoods. Computed correctly from the block fit, it is conservative in the simulation for β (0.96–0.98 with smooth errors and 100 curves, at about half of REML + CL2's width) and for Gaussian and Poisson responses, but it covers binary $\alpha(t)$ and $\gamma(t)$ at only 0.89. Over the combinations of truth and residual source on the six plasmode datasets with independent source curves, it undercovers β (0.87 on average; NCV + CL2 0.82, REML + CL2 0.93). Residual smoothing bias of the NCV estimate is a plausible contributor; variance and tail errors are not excluded (Supplement S4.2).
- *A smoothing-parameter uncertainty term for REML + CL2* raises coverage of β by 1.2–1.6 pp without improving the interval score (Supplement S4.3).
- *An AR(1) working model* (`bam()` with AR(1) errors restarting at each curve, correlation profiled by the restricted-likelihood criterion; Appendix B.1) covers 0.44–0.93 on the nine plasmode datasets (Supplement S3.1).
- *The curve bootstrap* (resampling curves and refitting) covers 0.93–0.97 on real residual curves, with intervals 1.4 times as wide as REML + CL2's and a median of 507 s per analyzed dataset (Supplement S3.2).
- *Penalized functional GEE* (Loewinger et al. 2026), in the R package fastFGEE (version 0.2.2). Under dependent errors, it improves the surface over REML by less than NCV (mean squared error 1.3–3.7 times NCV's). Wald intervals from its sandwich covariance cover $\beta(s,t)$ at only 0.67–0.89: their standard errors are well below the spread of the estimate (Supplement S3.3). Its interface offers a wild-bootstrap critical value for intercepts and scalar effects; we obtained it for the surface and fitted means from an internal routine, which is not a documented use of the package. With an independence working correlation and its t adjustment, these intervals for $\beta(s,t)$ were conservative under dependence in the simulation and slightly narrower than REML + CL2 with the Satterthwaite critical value (coverage 0.97–0.99 against 0.95–0.96, lower tail 0.94–0.98 against 0.92–0.94, at 0.88–0.99 times the width). On the 6 plasmode datasets with independent source curves (midpoint truth, REML residuals) and all subjects, they were again narrower (0.7–0.9 times) and covered 0.93–0.98 against 0.94–0.96, but on 5 of them their coverage was much less even over the grid (lower tail 0.75–0.85 against 0.89–0.92). The bootstrap calibrates one critical value for the whole surface, so it targets average coverage. With 40 subjects, the t adjustment often reached its floor of one degree of freedom and made the intervals 1.5–4.9 times as wide as those of REML + CL2 with the Satterthwaite critical value; without it, they covered as low as 0.89, against 0.94–0.96. Results for fitted means and for independent errors are in Supplement S3.3.
- *Clusters other than curves.* Block- and subject-clustered CL2 covered below nominal on the weather data, with 16 spatial blocks as clusters, and in a function-on-scalar design with 25 subjects and five curves per subject. With 100 or 500 subjects, subject-clustered CL2 was close to nominal in that

design (Supplement S2.3, S3.3).

- *A principal-component term for the covariate.* With the low-rank covariate of one synthetic setting, an `ffpc()` term with the 3 components that explain 95% of the covariate's variance estimated $\beta(s,t)$ about as well as NCV with `ff()` and much better than REML with `ff()`; its CL2 intervals with the normal critical value covered 0.91 (0.95 with five components; Supplement S1.9).

5 Plasmode study

5.1 Design

A plasmode study simulates from real data with a known truth (Franklin et al. 2014). Each of nine datasets has Gaussian responses and is fitted with the model in Equation 2 by REML and by curve-blocked NCV. A fit's fitted means serve as the truth: the *REML-fitted truth*, the *NCV-fitted truth*, and a *midpoint truth* refitted with every smoothing parameter at the geometric mean of the two fits' values. A replicate adds each subject's real residual curve, from the REML or the NCV fit, with a random sign (a *curve flip*) to the truth at its own covariates. This keeps the covariates, the residual shapes and their within-curve dependence, and makes the truth known. Because the signs are independent, the simulated errors are independent across curves, and each subject's error takes only two values, plus or minus its residual curve: the construction is a Rademacher wild cluster bootstrap around the fitted residuals, not a model of new real curves. Its truths are exactly representable by the fitted model, so it cannot evaluate misspecification of the mean. Two further limits follow from the construction. A curve's contribution to the sandwich varies little between replicates, so the plasmode study tests whether CL2's standard error matches the spread of the estimate, but hardly the Satterthwaite reference distribution, which accounts for the variability of that standard error; the evidence for the reference distribution is the simulation. Fitted residuals are also somewhat smaller than the errors that produced them, more so for a more flexible fit. A further error source for the ECG strain data, heartbeat difference curves, is not shaped by a fit (Appendix B.3).

In six datasets, the source curves are independent: ECG-derived myocardial strain, atrial strain from a randomized trial, knee moments during running and walking, the DTI tract profiles and an eight-lead ECG dataset. In three, the source curves are serially or spatially dependent (monthly ocean profiles, weather stations, weekly electricity demand); their simulated errors are still independent across curves, except in block-flip runs reported in the supplement. The midpoint fit lies between the REML and NCV fits for every term on only 3 of the 9 datasets, and its surface is rougher than both on running, ocean, weather; it is a third smoothing level, not a compromise. The datasets have 73–138 curves on 30–101 grid points, design effects of the residual curves (the factor by which within-curve correlation inflates the variance of a grid mean) of 6–40 and functional covariates of rank 4–24 (Appendix B.2). Each combination of dataset, truth and residual source is run with all subjects and with 40, with 200 replicates; the mean is scored for a fixed set of 50 subjects.

5.2 Results

Estimation. The median relative errors are 0.6–3.4 for REML and 0.5–0.8 for NCV across datasets (Figure 5, Figure 7). Weak identification of β in directions the functional covariate barely excites is a possible reason for their size (Section 1). Which fit is better depends on the truth: NCV's relative error is 0.07–0.87 times REML's under the NCV-fitted truth, 0.24–0.97 times under the midpoint truth and 0.55–1.69 times under the REML-fitted truth, where NCV is worse on 4 of the 9 datasets. A fit used as truth is shaped like the

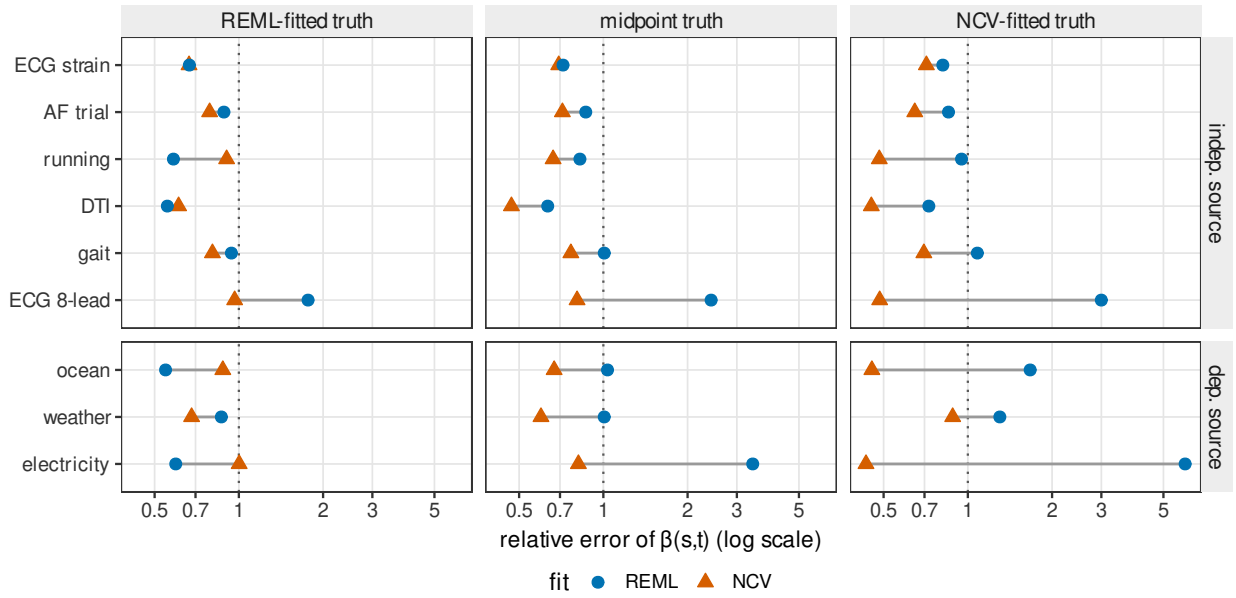


Figure 5: Plasmode relative error of the REML and NCV estimates of $\beta(s,t)$, all subjects, by dataset and truth source (median over replicates and the two residual sources; log scale; dotted: error as large as the effect).

selector that produced it, so the truth source favors that selector; the simulation, whose truths come from neither fit, is the cleaner comparison, although part of its gap reflects its smooth truths (Section 4.1). For the mean, NCV's relative error is 0.71–1.00 times REML's per dataset.

Intervals. With curve flips, the simulated errors are independent across curves on all nine datasets, so CL2's assumption holds on all of them. Model-based intervals cover 0.34–0.75 for the mean and β. REML + CL2 covers the mean at 0.92–0.95 on all nine datasets and β at 0.92–0.94 on eight (Figure 6), but β at only 0.88 on the weather data, which combine few curves (73), a functional covariate of rank 4 and rough residuals.

With the normal critical value, REML + CL2's coverage of β is below 0.95 in 130 of 130 combinations of dataset, truth, residual source, number of subjects and construction variant. Among the 120 combinations with curve flips, it is below 0.95 in 120 with the normal and in 63 with the Satterthwaite critical value. These combinations share data and are not independent confirmations. Its standard error is somewhat smaller than the replicate spread of the estimate (median ratio 0.89–1.01 per dataset), which is consistent with a standard error a little short, plus further error on some datasets. With 40 subjects on the six datasets with independent source curves, REML + CL2 covers β 1.4–4.8 pp below nominal and the mean 0.6–3.2 pp below; on the weather data, it covers β at 0.77–0.87.

With Satterthwaite degrees of freedom (Supplement S2.1), REML + CL2 covers β at 0.94 on average over the nine datasets with all subjects and 0.94 with 40 (normal critical value: 0.93 and 0.91), at 1.05 and 1.13 times the width. It covers the mean, $\alpha(t)$ and $\gamma(t)$ at 0.94–0.96. The weather surface stays short (0.90 with all subjects, 0.86 with 40). Adding the smoothing-parameter uncertainty term and using the t_{G-1} critical value helps about as much, with the weather surface again short (Supplement S4.3).

On the six datasets with independent source curves, NCV + CL2 covers β at 0.73–0.91, with lower tails 0.44–0.81, and loses up to 19 pp with 40 subjects. Its standard error is short (median ratio to the replicate standard deviation 0.75–0.98 across these datasets). A refit with the smoothing parameters fixed at their median restores it (1.03–1.15), but coverage then still stays at 0.69–0.92. Fixing the smoothing parameters changes

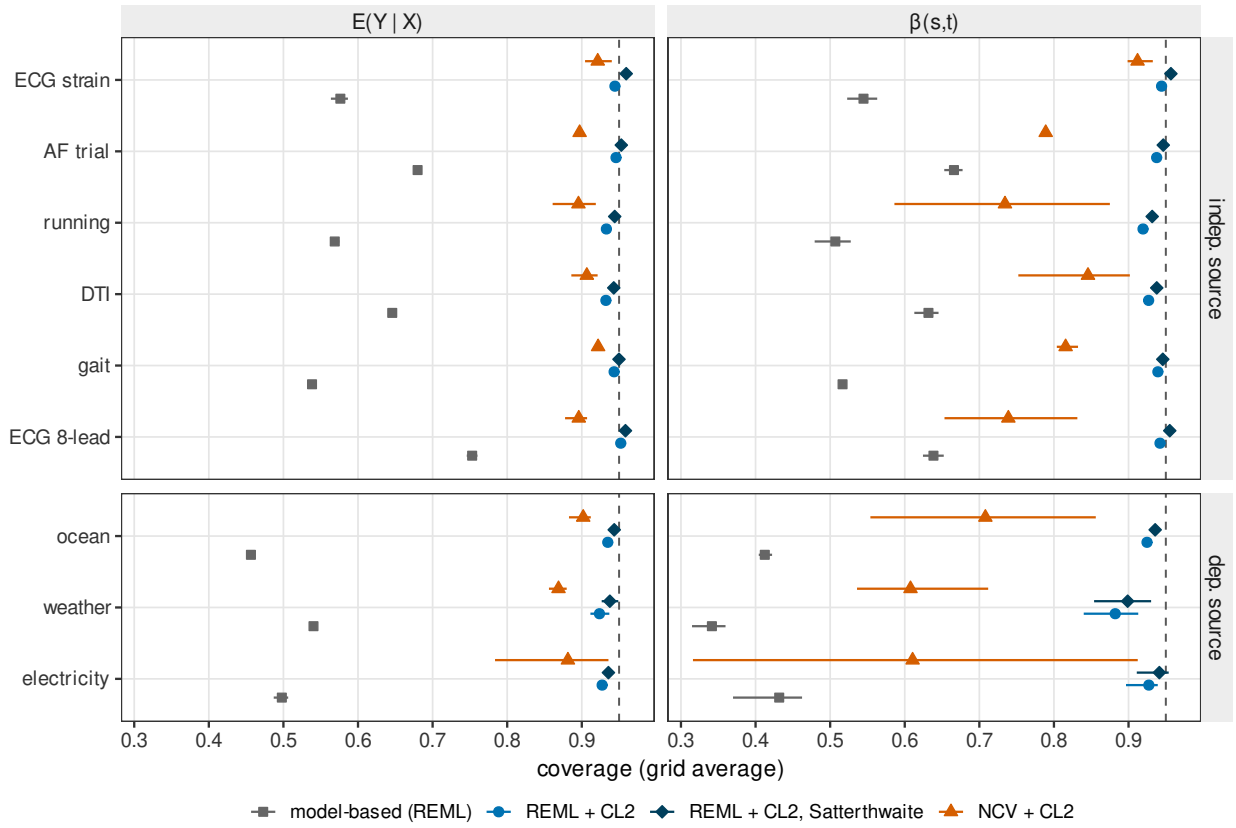


Figure 6: Plasmode coverage of nominal 95% pointwise intervals for the mean and $\beta(s,t)$, all subjects: mean over truth and residual sources (symbols) and their range (bars). REML + CL2 and NCV + CL2 use the normal critical value. Dashed: 0.95.

both the amount and the variability of smoothing, so this is a diagnostic, not an exact decomposition; it suggests smoothing bias as the larger part (Supplement S4.1).

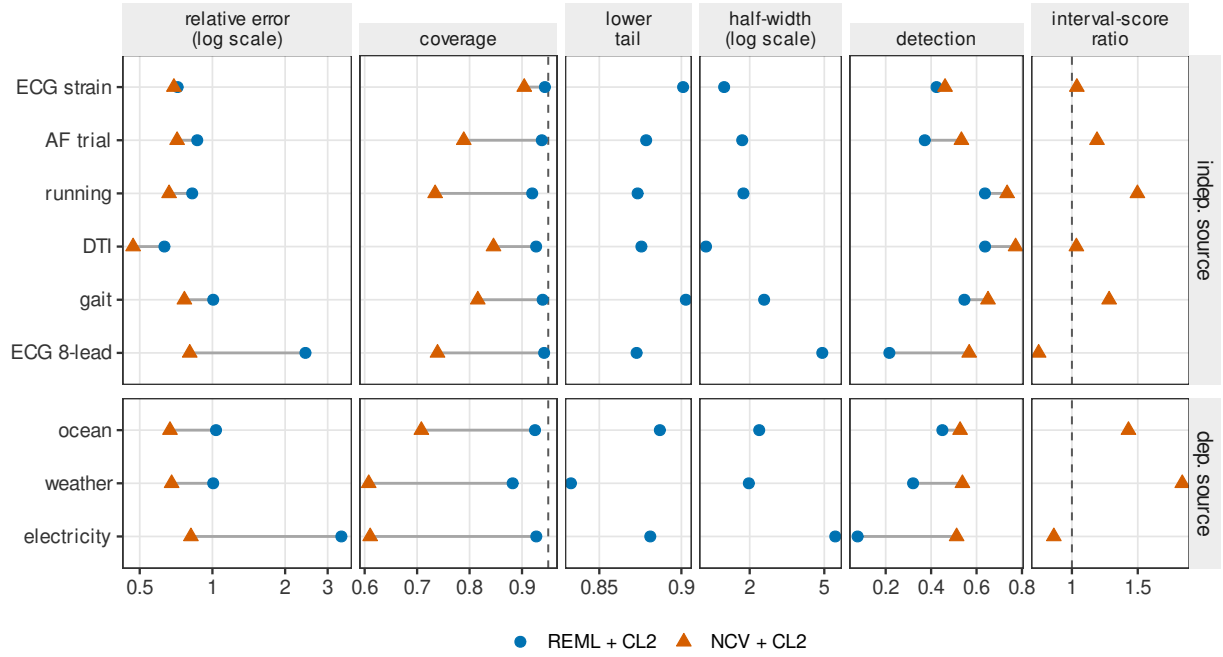


Figure 7: Plasmode study, $\beta(s,t)$, all subjects, per dataset: median relative error of the REML and NCV estimates; coverage, detection rate; for REML + CL2 the lower tail (5% quantile of pointwise coverage) and the relative half-width; for NCV + CL2 the interval score relative to REML + CL2 (means, medians or geometric means over truth and residual sources). Gray segments join the two intervals of a dataset; log scales for relative error and half-width. Dashed: 0.95 for coverage, 1 for the ratio. Values: Supplement S7.

Informativeness. As in the simulation, REML + CL2 is close to nominal but wide for β: its half-width is 1.2–5.7 times the effect, and it detects 0.08–0.64 of the grid points where the true value exceeds 25% of its maximum. NCV + CL2 detects more (0.46–0.77) but does not hold its level, and here its interval score is 0.7–1.8 times REML + CL2's (higher on 7 of the 9 datasets).

Intervals centered at the NCV estimate with a bias correction and a smoothing-parameter term (Section 4.5) cover β at 0.83–0.94 on the six datasets with independent source curves, against 0.91–0.94 for REML + CL2, at 0.7–1.1 times its width and with lower tails as low as 0.38 in single combinations. On the three datasets with dependent source curves, they cover 0.66–0.84 (Supplement S4.1). These comparisons are descriptive: the NCV-centered intervals were computed on 50 replicates and REML + CL2 on 100, without a paired measure of precision.

The NCV estimate with REML + CL2's half-width covers 0.98–1.00 in the simulation (dependent errors, 100 curves, smooth and rough truths) and 0.97 and 0.96 under the NCV-fitted and the midpoint plasmode truth (all subjects, mean over datasets). Under the REML-fitted truth, it covers 0.91 with a lower tail of 0.67, against a lower tail of 0.87 for REML + CL2. It is no narrower than REML + CL2 and detects less on all nine datasets (Supplement S4.1).

6 Recommendations and software

Recommendations for applied work with pffr()

1. **Assume within-curve dependence.** Do not use model-based intervals of the working-independence fit, and expect REML to undersmooth coefficient surfaces.
2. **Shape of a surface: curve-blocked NCV.** Fit the model with `method = "NCV"` and show its surface estimate as the description of the effect.
3. **Inference: REML + CL2 with Satterthwaite degrees of freedom**, for all terms and for fitted means. The intervals for a surface are wide; report them as they are, and as pointwise intervals.
4. **Both goals: fit twice.** Take the surface from the NCV fit and the intervals from the REML fit, and show them side by side, not as a band around the NCV surface.
5. **Means, intercepts and scalar effects only:** one REML fit with these intervals is enough, with the exceptions in 6.
6. **Treat as approximate:** intervals for intercepts and scalar effects of binary responses and for misregistered counts, and any interval with fewer than 20 curves or with few clusters of curves.
7. **Check the basis.** Refit with a richer basis and compare; do not interpret the shape of a surface that changes with the basis.

The evidence for each recommendation and its limits are as follows.

Dependence (1). Under dependence, REML undersmooths coefficient surfaces (Section 4.1), and model-based intervals undercover, more so on dense grids (Section 4.2).

Shape of a surface (2). The forthcoming refund releases set one neighborhood per curve (per `cluster` if given) for `method = "NCV"`; in earlier releases and in plain `gam()`, `method = "NCV"` alone leaves out single points, and the neighborhoods must be supplied through `nei`. NCV's gain in estimation also holds, more weakly, for smooth effects of scalar covariates that vary along t (Section 4.1). When the surface was zero, the NCV estimate stayed much closer to zero than REML's, which showed structure (Section 4.4). NCV is not uniformly better: it was worse on some plasmode datasets under the REML-fitted truth, which favors REML (Section 5.2). Apart from misregistration, a small share of its fits under dependence have an unusually small smoothing parameter and a larger error, which convergence flags rarely reveal outside scaled-t responses, although NCV still estimated the surface better than REML on nearly all of these replicates (Section 4.1). For Gaussian responses, pcre and GLS with an FPCA residual covariance also estimated the surface much better than REML, but worse than NCV in the simulation and comparably to it in a plasmode comparison (Section 4.1). If mgcv flags an NCV fit as not converged, as it usually did for scaled-t responses, refitting from other starting smoothing parameters and comparing the criterion and the surface is a simple check (Appendix B.1); it was not evaluated as a procedure.

Inference (3). The recommended interval uses the curve-clustered CL2 covariance of the REML fit with the full block and the Satterthwaite critical value. It was evaluated for all four estimands in the main factorial at the default signal with 20, 40 and 100 curves, in all extensions and in the plasmode study. Under dependence, it was close to nominal for the mean, $\beta(s,t)$ and $\gamma(t)$ in the main factorial and for $\beta(s,t)$ on average in the plasmode study, but not on one dataset with few curves and a low-rank covariate (Section 4.2, Section 5.2). Under independent errors, it was conservative for $\beta(s,t)$ (Section 4.2). The price is width, which is largest with few curves. fastFGEE's wild-bootstrap intervals for $\beta(s,t)$, obtained from an internal routine, were conservative and slightly narrower under dependence in the simulation, but we keep REML + CL2 because on most of the six plasmode datasets with independent source curves its pointwise coverage was more even, and it kept its coverage at 40 subjects without the large widening of fastFGEE's t adjustment (Section 4.5).

Only pointwise intervals were evaluated: a pointwise band answers "is there an effect, and where" poorly (Section 5.2) and "is $\beta(s,t)$ zero everywhere" not at all, which needs simultaneous bands or global tests (Section 4.4).

Fitting twice (4, 5). The two fits differ only in how the smoothing parameters are chosen; Section 4.3 explains why they differ. In our Gaussian benchmark with 100 curves, an NCV fit took about 4 times as long as a REML fit (Appendix C). Disregard the NCV fit's standard errors: NCV + CL2 intervals undercover (Section 4.3, Section 5.2). One way to report both fits is the NCV surface next to REML + CL2's map of grid points whose interval excludes zero, labeled as pointwise. Under a zero surface, most replicates show some excluded points (Section 4.4); this display and disagreements between the two fits were not evaluated. In the dependent main-factorial settings at the default signal, the NCV estimate lay outside the Satterthwaite interval of REML + CL2 at 1.2–2.4% of the grid points. Centering REML + CL2's half-width at the NCV estimate is not a shortcut to a band around the NCV surface: it overcovers in the simulation, covers unevenly under the REML-fitted plasmode truth, and detects less than REML + CL2 (Section 5.2). For the functional intercept, scalar effects and fitted means, REML + CL2 gives both the estimate and the intervals, with the exceptions that follow.

Approximate intervals (6). Intervals for the intercept and scalar effects of binary responses and for misregistered counts undercover, also with Satterthwaite degrees of freedom (Section 4.2). With 20 curves and dependent errors in the main factorial, the Satterthwaite critical value brought the mean, $\beta(s,t)$ and $\gamma(t)$ close to nominal, but not binary intercepts; with 20 curves and independent errors, every interval undercovered binary intercepts and scalar effects (Section 4.2). Nothing below 20 curves, or with 20 curves outside the main factorial, was evaluated; treat intervals there as approximate. With several curves per subject, cluster by subject (`cluster =`). Subject clustering leaves fewer clusters and does not address dependence between subjects; block- and subject-clustered CL2 covered below nominal with 16 blocks and with 25 subjects, and close to nominal with 100 or 500 subjects (Section 4.5). With `method = "NCV"`, the same argument gives subject neighborhoods, which were not evaluated.

Basis and identification (7). A richer basis protects against approximation bias but widens REML + CL2's intervals (Section 4.3). It does not help where the functional covariate carries little information about a direction of the surface: the surface is then weakly identified (Scheipl and Greven 2016). Refrain from interpreting the surface's shape when the estimate or its interval changes substantially with the basis, or when the covariate's principal components leave the relevant directions poorly covered given the penalty's null space; agreement between two penalized fits does not certify identification. Fitted means or well-supported contrasts are then safer targets, though not protected against misspecification (Supplement S5). If the functional covariate's variation is concentrated in a few principal components, an `ffpc()` term with a small number of components can replace `ff()` (Section 4.5). For covariates with a richer spectrum, keep `ff()`: few components truncate the surface and the intervals undercover, and a high variance-explained threshold, such as 0.99 (the default of current refund releases), gives unstable, very noisy estimates, as is known for functional principal component regression, an ill-posed problem (Hall and Horowitz 2007; Reiss and Ogden 2007).

Software. The study used refund at development commit 79a346fb, which contains these procedures, and an unpatched and a patched pre-release build of mgcv 1.9-5; the patch holds the corrections found in this study, among them those of mgcv's covariances for NCV fits (Supplement S4.2, S6). Forthcoming releases of refund and mgcv will implement these recommendations, with REML + CL2 and Satterthwaite degrees of freedom as the default intervals of `pffr()` fits. Supplement S6 gives example code, the software used in the study and a reproduction route.

7 Discussion

Under within-curve dependence, the working-independence REML fit of a penalized function-on-function regression gives an undersmoothed, noisy estimate of the coefficient surface and model-based intervals that are far too narrow. Curve-blocked NCV, which typically smooths more, improves the estimate but not the intervals. A curve-clustered sandwich with a leverage adjustment for the penalized hat and Satterthwaite degrees of freedom brings the intervals close to nominal for most estimands of Gaussian, count and binary responses in the simulation, down to 20 curves, and for the surface on eight of the nine plasmode datasets, but it cannot improve the estimate it is centered on. The two remedies thus show the undersmoothing-versus-bias-correction tension of nonparametric inference (Hall 1992; Calonico, Cattaneo, and Farrell 2018), here produced by the dependence, hence our recommendation to fit twice when the surface matters. Intermediate amounts of smoothing between REML and NCV, global tests of $\beta \equiv 0$ and simultaneous bands are the natural next steps, as is a wild cluster bootstrap critical value (Cameron, Gelbach, and Miller 2008) for REML + CL2 with few curves; they were not studied here. The evidence has the following limits.

- *Binary responses.* REML + CL2 undercovers intercepts and scalar effects of binary responses, also under independent errors and with Satterthwaite degrees of freedom. The diagnostics are compatible with a standard error that is too small for these terms, partly through smoothing-parameter variability; local bias is not excluded. Every interval centered at the NCV estimate undercovers for binary responses (Supplement S4.1).
- *Misregistration.* When curves are observed on their own clocks, the observed-clock mean of a count response is not a generalized linear predictor, and no method covers it; for Gaussian responses, REML + CL2 is unaffected.
- *Design.* The synthetic function-on-function design has 20 curves only in the main factorial and at most 100 curves, a zero surface only for Gaussian and binary responses with independent and smooth errors, one empirically derived smooth error kernel (from the DTI data) beside parametric kernels, no dependent error process of intermediate strength, regular grids without missing values, and first-order difference penalties throughout. For count and binary responses, dependence enters only through a Gaussian copula.
- *Mean model.* The study addresses misspecified dependence under a correct model for the mean, and all synthetic truths in the main study are projected onto the fitted bases. For the smooth $\beta(s,t)$ of the Gaussian default setting, the projection is immaterial: scored against the exact function, REML + CL2 covers it at 0.93–0.94 under smooth errors (Supplement S5.1, normal critical value). A basis that cannot represent the effect is a misspecified mean model, and none of the intervals studied here protects against it (Supplement S5.1). With a local bump of standard deviation 0.05 added to the surface, REML + CL2 covers $\beta(s,t)$ at only 0.17–0.58 inside the bump. For a truth with components the covariates barely excite and the default basis cannot represent, it covers $\beta(s,t)$ at only 0.44–0.56 under smooth errors and 0.30 under independent errors. A basis with four times finer knot spacing restores 0.93 under smooth errors, at the price of wider intervals, but only 0.86 under independent errors; NCV + CL2 stays short even then. These coverages refer to the exact truth, which mixes approximation bias with weak identification.
- *Plasmode study.* Fits used as truths favor their own selector, the errors are sign-flipped fitted residuals with independent signs across curves, the truths are representable by the model, and all nine datasets are Gaussian and were chosen, not sampled. With sign flips, the plasmode results test CL2's standard error, but hardly the Satterthwaite reference distribution (Section 5.1).
- *Intervals around the NCV estimate.* Their shortfall is mainly smoothing bias in the simulation; on the plasmode data, the diagnostics suggest smoothing bias and the variability of the selected smoothing parameters, but fixing the smoothing parameters changes both. Neither a one-step bias correction

nor a variance term for the smoothing parameters, nor their combination as a frozen-map approximation, gave a narrower calibrated interval in the plasmode study. Calibrated intervals after data-driven heavier smoothing remain open for the NCV fit. A wild-bootstrap critical value around fastFGEE's fit gave conservative intervals for β, narrower than REML + CL2 with the Satterthwaite critical value, under dependence in the simulation. On most of the plasmode datasets with independent source curves, its coverage was less even with all subjects, and with 40 subjects its intervals were very wide with the t adjustment and undercovered without it. That fit's mean squared error for β under dependence in the simulation is still 1.3–3.7 times NCV's (Section 4.5).

Code and data availability

Code and summary files are at https://github.com/fabian-s/fof-inf; the tables and figures of this paper and its supplement are computed from those summary files. The per-replicate records are at https://doi.org/10.5281/zenodo.23916388. Records of the two cardiology datasets are patient and volunteer data and are not shared; only aggregate results are available for them. The other datasets are public (Appendix B.2).

Use of generative AI

All code and all text of this paper and its supplement were generated by large language models: Claude Opus 5.5 and Claude Fable 5.1 (Anthropic) and ChatGPT models Sol and Astra, versions 5.6 and 6 (OpenAI). The study design, structural decisions and research directions were developed collaboratively between the author and these models. The author takes full responsibility for the content.

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A The CL2 covariance

Notation follows the main text: $\mathbf{W}$ holds the working weights $w_{ij} = (d\mu/d\eta)^2_{ij}/\{\phi V(\mu_{ij})\}$, $\mathbf{S} = \sum_r (\lambda_r/\phi)\mathbf{S}_r$, $\mathbf{V}_p = (\mathbf{X}^\top\mathbf{W}\mathbf{X} + \mathbf{S})^{-1}$, $\mathbf{V}_e = \mathbf{V}_p\mathbf{X}^\top\mathbf{W}\mathbf{X}\mathbf{V}_p$, and for curve i $\tilde{\mathbf{X}}_i = \mathbf{W}_i^{1/2}\mathbf{X}_i$ and $z_{ij} = (y_{ij} - \hat{\mu}_{ij})/\{\phi V(\hat{\mu}_{ij})\}^{1/2}$. The stacked whitened design is $\tilde{\mathbf{X}}$, and $\tilde{\mathbf{H}} = \tilde{\mathbf{X}}\mathbf{V}_p\tilde{\mathbf{X}}^\top$ has blocks $\tilde{\mathbf{H}}_{ih}$.

A.1 Scores and the penalized bread

For an exponential-family response, the score of the log-likelihood contribution of y_{ij} is

$$\mathbf{s}_{ij}(\theta) = \frac{(y_{ij} - \mu_{ij})\,(d\mu/d\eta)_{ij}}{\phi V(\mu_{ij})}\,\mathbf{x}_{ij} = z_{ij}\,\tilde{\mathbf{x}}_{ij}, \qquad \tilde{\mathbf{x}}_{ij} = w_{ij}^{1/2}\mathbf{x}_{ij}, \tag{6}$$

so the score of curve i is $\mathbf{u}_i = \sum_j \mathbf{s}_{ij} = \tilde{\mathbf{X}}_i^\top \mathbf{z}_i$. The fit maximizes $\sum_{i,j} \ell_{ij}(\theta) - \frac{1}{2}\theta^\top \mathrm{S}\theta$, whose estimating equation is

$$\sum_{i=1}^{G} \mathbf{u}_i(\hat{\theta}) = \mathrm{S}\hat{\theta}. \tag{7}$$

The negative Hessian of the penalized log-likelihood is $\mathbf{X}^\top\mathbf{W}\mathbf{X} + \mathrm{S} = \mathbf{V}_p^{-1}$ (exactly for canonical links, in expectation otherwise). Linearizing Equation 7 around the target θ^* at which its expectation vanishes gives

$$\hat{\theta} - \theta^* \approx \mathbf{V}_p \sum_{i=1}^{G} \{\mathbf{u}_i(\theta^*) - \mathrm{E}\,\mathbf{u}_i(\theta^*)\}, \qquad \mathrm{Var}(\hat{\theta}) \approx \mathbf{V}_p \Big\{\sum_{i=1}^{G} \mathrm{Var}\,\mathbf{u}_i(\theta^*)\Big\}\mathbf{V}_p, \tag{8}$$

so $\mathbf{V}_p$ is the bread and only the meat $\sum_i \mathrm{Var}\,\mathbf{u}_i$ depends on the within-curve dependence. Under the working model, $\mathrm{Var}(\mathbf{z}_i) = \mathbf{I}$ and the meat is $\tilde{\mathbf{X}}^\top\tilde{\mathbf{X}}$, which returns $\mathbf{V}_e$. The unpenalized bread $(\mathbf{X}^\top\mathbf{W}\mathbf{X})^{-1}$ linearizes a different estimator. Scores and penalty are both on the scale of the summed log-likelihood; a construction that averages the scores over the N observations must use the penalty $\sum_r \lambda_r \mathrm{S}_r/(N\phi)$, or the bread and the shrinkage are mis-scaled by the factor N (and by ϕ if it is dropped).

By Equation 7, the curve scores at the estimate do not sum to zero: $\sum_i \tilde{\mathbf{X}}_i^\top \mathbf{z}_i = \mathrm{S}\hat{\theta}$. Unlike the unpenalized cluster sandwich (Liang and Zeger 1986), whose curve scores at the estimate sum to zero, the meat $\sum_i \mathbf{U}_i\mathbf{U}_i^\top$ below is not centered; Proposition A.1(b) gives the consequence.

A.2 The leverage adjustment

The CL2 covariance of the main text is

$$\hat{\mathbf{V}}_{\mathrm{CL2}} = \frac{G}{G-1}\,\mathbf{V}_p\Big(\sum_{i=1}^{G} \mathbf{U}_i\mathbf{U}_i^\top\Big)\mathbf{V}_p + (\mathbf{V}_p - \mathbf{V}_e), \qquad \mathbf{U}_i = \tilde{\mathbf{X}}_i^\top \mathbf{R}_i^{-1/2}\mathbf{z}_i, \tag{9}$$

with $\mathbf{R}_i = [(\mathbf{I} - \tilde{\mathbf{H}})(\mathbf{I} - \tilde{\mathbf{H}})^\top]_{ii}$. Since $\tilde{\mathbf{X}}^\top\tilde{\mathbf{X}} = \mathbf{V}_p^{-1} - \mathrm{S}$, $(\mathbf{I} - \tilde{\mathbf{H}})^2 = \mathbf{I} - \tilde{\mathbf{X}}(2\mathbf{V}_p - \mathbf{V}_p\tilde{\mathbf{X}}^\top\tilde{\mathbf{X}}\mathbf{V}_p)\tilde{\mathbf{X}}^\top$ and

$$\mathbf{R}_i = \mathbf{I} - \tilde{\mathbf{X}}_i\big(2\mathbf{V}_p - \mathbf{V}_p\tilde{\mathbf{X}}^\top\tilde{\mathbf{X}}\mathbf{V}_p\big)\tilde{\mathbf{X}}_i^\top = \mathbf{I} - \tilde{\mathbf{H}}_{ii} - \tilde{\mathbf{X}}_i\mathbf{V}_p\mathrm{S}\mathbf{V}_p\tilde{\mathbf{X}}_i^\top. \tag{10}$$

$\mathbf{R}_i$ needs only $\tilde{\mathbf{X}}_i$ and one $p \times p$ matrix. It is computed in the column space of $\tilde{\mathbf{X}}_i$ in $\mathbb{R}^{D_i}$ (dimension at most $\min(D_i, p)$), because $\mathbf{R}_i$ equals $\mathbf{I}$ on its orthogonal complement and $\tilde{\mathbf{X}}_i^\top$ annihilates that complement.

Relation to CR2. The bias-reduced linearization of Bell and McCaffrey (2002) chooses $\mathbf{A}_i$ with $\mathbf{A}_i[(\mathbf{I}-\mathbf{H})\Phi(\mathbf{I}-\mathbf{H})^\top]_{ii}\mathbf{A}_i = \Phi_i$ for a working covariance Φ (McCaffrey and Bell 2006; Pustejovsky and Tipton 2018). With $\Phi = \mathbf{I}$ this is $\mathbf{A}_i = \mathbf{R}_i^{-1/2}$. For a projection, $\mathbf{R}_i = \mathbf{I} - \mathbf{H}_{ii}$ and the usual CR2 results. The penalized hat is not idempotent: the last term of Equation 10 is positive semidefinite, so $\mathbf{R}_i \preceq \mathbf{I} - \tilde{\mathbf{H}}_{ii}$, and the approximation $\mathbf{A}_i^{\mathrm{s}} = (\mathbf{I} - \tilde{\mathbf{H}}_{ii})^{-1/2}$, which keeps the full block $\tilde{\mathbf{H}}_{ii}$ but drops the last term of Equation 10 (the *shortcut*), under-corrects: $\mathbf{A}_i^{\mathrm{s}}\mathbf{R}_i\mathbf{A}_i^{\mathrm{s}} \preceq \mathbf{I}$. The dropped term is small when the per-curve leverage is small.

Expectation of the meat. Take the Gaussian working model with fixed λ and fixed design: $\mathbf{W}^{1/2}\mathbf{y} = \tilde{\mathbf{X}}\theta + \varepsilon$ with $\mathrm{Var}(\varepsilon) = \mathbf{I}$. Then $\mathbf{z} = (\mathbf{I} - \tilde{\mathbf{H}})(\tilde{\mathbf{X}}\theta + \varepsilon)$, and $(\mathbf{I} - \tilde{\mathbf{H}})\tilde{\mathbf{X}} = \tilde{\mathbf{X}}\mathbf{V}_p\mathbf{S}$ gives

$$\mathrm{E}(\mathbf{z}_i) = \mathbf{m}_i = \tilde{\mathbf{X}}_i\mathbf{V}_p\mathbf{S}\theta, \qquad \mathrm{E}(\mathbf{z}_i\mathbf{z}_i^\top) = \mathbf{R}_i + \mathbf{m}_i\mathbf{m}_i^\top. \tag{11}$$

The penalized residuals have nonzero mean unless $\mathbf{S}\theta = \mathbf{0}$. With $\mathbf{R}_i$ invertible and the floor inactive,

$$\mathrm{E}\Big(\sum_i \mathbf{U}_i\mathbf{U}_i^\top\Big) = \sum_i \tilde{\mathbf{X}}_i^\top\tilde{\mathbf{X}}_i + \sum_i \tilde{\mathbf{X}}_i^\top\mathbf{R}_i^{-1/2}\mathbf{m}_i\mathbf{m}_i^\top\mathbf{R}_i^{-1/2}\tilde{\mathbf{X}}_i. \tag{12}$$

The first term is the working-model sampling meat, $\mathrm{Var}(\tilde{\mathbf{X}}^\top\varepsilon)$, so $\mathbf{V}_p(\sum_i \mathbf{U}_i\mathbf{U}_i^\top)\mathbf{V}_p$ is unbiased for $\mathbf{V}_e$ when the truth lies in the null space of the penalty and biased upwards by a positive semidefinite term otherwise. The factor $G/(G-1)$ adds a further upward bias. Under within-curve dependence, $\mathrm{Var}(\varepsilon_i) = \Sigma_i \neq \mathbf{I}$, and the adjustment removes the residual shrinkage only approximately; it does not whiten correlated residuals.

Floor and factor. The eigenvalues of $\mathbf{R}_i$ are floored at 10^{-6} before the inverse square root, which caps the inflation of any residual direction at a factor 10^3. The floor binds only for curves whose data nearly determine a coefficient direction on their own. How often it bound in the REML and NCV fits of the two studies was not recorded. In the comparator runs (Section 4.3), it bound in no GLS fit and, among the pcre fits, only on the running data, there in 70 of 100 replicates with all subjects and in 60 of 100 with 40 subjects, so pcre + CL2 on that dataset depends on the floor. CR2 carries no factor $G/(G-1)$; we keep it. The factor and the uncentered scores of Appendix A.1 are each conservative at order $1/G$. At the estimate, the curve scores sum to $\mathbf{S}\hat{\theta}$, so the uncentered meat adds about $\mathbf{V}_p\mathbf{S}\hat{\theta}\hat{\theta}^\top\mathbf{S}\mathbf{V}_p/G$ to the covariance, a plug-in for the squared smoothing bias divided by G. We keep both because, under dependence, the leverage adjustment removes the residual shrinkage only approximately and CL2's standard error is short even with them (Appendix A.6); their separate contributions were not evaluated. The factor inflates frequentist CL2 standard errors by 1.3% at $G = 40$ and 0.5% at $G = 100$; in the Bayesian form it multiplies only the sandwich part, so these are upper bounds.

CR1 is Equation 9 with $\mathbf{R}_i = \mathbf{I}$ and no further degrees-of-freedom factor.

A.3 Bayesian and frequentist forms

The frequentist form omits $\mathbf{V}_p - \mathbf{V}_e = \mathbf{V}_p\mathbf{S}\mathbf{V}_p$. For Gaussian responses with identity link and fixed λ, the bias of $\hat{\theta}$ is $-\mathbf{V}_p\mathbf{S}\theta$. Under the smoothing prior $\theta \sim N(\mathbf{0}, \mathbf{S}^-)$, $\mathrm{E}(\mathbf{V}_p\mathbf{S}\theta\theta^\top\mathbf{S}\mathbf{V}_p) = \mathbf{V}_p\mathbf{S}\mathbf{V}_p$: the allowance is the prior-averaged squared bias, and $\mathbf{V}_p$ the prior-averaged mean squared error matrix (Wahba 1983; Nychka 1988; Marra and Wood 2012). For a fixed truth and any contrast $\mathbf{a}$, by the Cauchy–Schwarz inequality,

$$(\mathbf{a}^\top\mathbf{V}_p\mathbf{S}\theta)^2 \le (\mathbf{a}^\top\mathbf{V}_p\mathbf{S}\mathbf{V}_p\mathbf{a})\,(\theta^\top\mathbf{S}\theta), \qquad \theta^\top\mathbf{S}\theta = \sum_r (\lambda_r/\phi)\,\theta^\top\mathbf{S}_r\theta. \tag{13}$$

The allowance bounds the squared bias of every contrast if $\theta^\top\mathbf{S}\theta \le 1$; for larger values the bound permits, but does not imply, a bias above the allowance.

Where the data identify θ ($\tilde{\mathbf{X}}^\top\tilde{\mathbf{X}}$ positive definite), the squared bias scales as λ^2 and the allowance as λ for small λ. Undersmoothing, as REML produces under within-curve dependence, therefore makes the allowance large relative to the bias but small relative to the dependence-inflated variance, and the two forms nearly coincide. For a truth that is rough at the selected λ ($\theta^\top\mathbf{S}\theta$ large), or after oversmoothing, the bias can exceed the allowance, most likely where the truth is most curved. Under independent errors with a smooth truth the allowance exceeds the bias and the Bayesian form overcovers.

A.4 Extended families

The CL2 construction needs a factorization $\mathbf{s}_{ij} = z_{ij}\tilde{\mathbf{x}}_{ij}$ in which $\tilde{\mathbf{x}}_{ij}$ carries the Fisher weight of the fit, so that $\operatorname{tr}\tilde{\mathbf{H}}$ equals the effective degrees of freedom.

Gaussian location–scale (mgcv's gaulss). The two linear predictors η_1 (mean) and η_2 (inverse standard deviation τ) have block-diagonal Fisher information with weights $W_1 = \tau^2(d\mu/d\eta_1)^2$ and $W_2 = 2(d\tau/d\eta_2)^2/\tau^2$. Each observation contributes two pseudo-rows, both assigned to its curve,

$$\tilde{\mathbf{x}}_k = W_k^{1/2}\mathbf{x}_k, \qquad z_k = \frac{\partial\ell/\partial\eta_k}{W_k^{1/2}}, \qquad k = 1, 2, \tag{14}$$

with $\partial\ell/\partial\eta_1 = \tau^2(y-\mu)\,d\mu/d\eta_1$ and $\partial\ell/\partial\eta_2 = (1/\tau - \tau(y-\mu)^2)\,d\tau/d\eta_2$; $\mathbf{x}_k$ is the design row of predictor k, zero in the other block.

Scaled t (scat). With degrees of freedom ν and scale σ at their estimates, the exact location score and Fisher weight are

$$\frac{\partial\ell}{\partial\mu} = \frac{(\nu+1)(y-\mu)}{\nu\sigma^2 + (y-\mu)^2}, \qquad W = \Big(\frac{d\mu}{d\eta}\Big)^2 \frac{\nu+1}{(\nu+3)\sigma^2}, \tag{15}$$

and $z = (\partial\ell/\partial\mu)(d\mu/d\eta)/W^{1/2}$. The working-residual score Equation 6 would treat the t response as Gaussian and give no downweighting of outliers.

All other extended families of mgcv (negative binomial, Tweedie, beta, ordered categorical) use the working-residual score Equation 6 with the fitted variance function. For the negative binomial, with its shape parameter at the estimate, this is the exact score for the mean; for the beta family it is an approximation. The simulation uses CL2 for the beta and negative-binomial responses. Families that define their own sandwich (multinomial) have no cluster-robust covariance.

A.5 Asymptotic validity for fixed smoothing parameters

Proposition A.1. *Assume (i) the curves $(\mathbf{y}_i, \mathbf{X}_i)$, $i = 1, \dots, G$, are independent and identically distributed with $D_i \le \bar{D} < \infty$, with arbitrary dependence within a curve; (ii) the basis, hence p, is fixed and the link is canonical; (iii) $\lambda = \lambda_G$ is non-random with $\mathbf{S}_G/G \to \mathbf{S}_\infty$; (iv) the weighted design rows are uniformly bounded, the curve scores have bounded moments of order $2+\delta$ and dominated derivatives near the target, and $\mathbf{J} = \mathrm{E}(\tilde{\mathbf{X}}_1^\top\tilde{\mathbf{X}}_1) + \mathbf{S}_\infty$ is positive definite at $\theta^*_\infty = \lim_G \theta^*_G$, where θ^*_G solves*

$$\mathrm{E}\,\mathbf{u}_1(\theta) = (\mathbf{S}_G/G)\,\theta. \tag{16}$$

Then (a) $\sqrt{G}(\hat\theta - \theta^*_G) \to N(\mathbf{0}, \mathbf{J}^{-1}\mathbf{M}\mathbf{J}^{-1})$ with $\mathbf{M} = \operatorname{Var}\mathbf{u}_1(\theta^*_\infty)$. (b) With $\mu = \mathrm{E}\,\mathbf{u}_1(\theta^*_\infty) = \mathbf{S}_\infty\theta^*_\infty$, the variance part of Equation 9 (and of CR1) satisfies

$$G\,\frac{G}{G-1}\,\mathbf{V}_p\Big(\sum_i \mathbf{U}_i\mathbf{U}_i^\top\Big)\mathbf{V}_p \;\to_p\; \mathbf{J}^{-1}(\mathbf{M} + \mu\mu^\top)\mathbf{J}^{-1}, \qquad G(\mathbf{V}_p - \mathbf{V}_e) \to \mathbf{J}^{-1}\mathbf{S}_\infty\mathbf{J}^{-1}, \tag{17}$$

so the variance part is consistent for the covariance in (a) if $\mathbf{S}_\infty\theta^*_\infty = \mathbf{0}$ and conservative otherwise; the Bayesian form adds $\mathbf{J}^{-1}\mathbf{S}_\infty\mathbf{J}^{-1}$. (c) If the mean model is correct and $\sqrt{G}(\theta^*_G - \theta) \to \mathbf{0}$ (which forces $\mathbf{S}_\infty\theta = \mathbf{0}$ even if $\mathbf{S}_\infty \ne \mathbf{0}$), the frequentist CL2 interval for any fixed contrast is asymptotically valid for the true θ, and the Bayesian one is valid and conservative, asymptotically exact if $\mathbf{S}_\infty = \mathbf{0}$, provided the limiting variance $\mathbf{a}^\top\mathbf{J}^{-1}\mathbf{M}\mathbf{J}^{-1}\mathbf{a}$ of the contrast is positive.

The weights in $\tilde{\mathbf{X}}_1$ are evaluated at θ^*_∞, and a dispersion or other nuisance parameter of the family is taken as known. The covariance in (a) is unconditional: with random covariates, $\mathbf{M}$ includes the variation over curves of the conditional mean of the score, which is not zero when θ^*_∞ differs from θ. Appendix A.2 instead conditions on the design, where the same term appears as an upward bias of the meat.

For Gaussian responses with identity link, the finite-sample target given the design is $\mathrm{E}(\hat{\theta}) = (\mathbf{I} - \mathbf{V}_p\mathbf{S})\theta$, of which Equation 16 is the population version; for GLMs the target solves the expected penalized score equation.

Proof. $\hat{\theta}$ solves an M-estimating equation with independent curve contributions and a deterministic penalty offset whose expectation vanishes at θ^*_G; (a) follows from the standard argument for such equations (Liang and Zeger 1986; Hansen and Lee 2019). For (b), bounded curve sizes and fixed p make the leverages $O_p(G^{-1})$, so CL2 and CR1 meats agree to first order, and the meat converges to the second moment of a score with mean μ. For (c), a correct mean model gives curve scores with mean zero whatever the dependence within the curve. □

The proposition does not cover smoothing parameters selected from the data, nor a number of coefficients comparable to or larger than G; both hold in our applications and simulations.

A.6 Covariance components

The comparison uses the 18 main factorial settings at the default signal level ($D = 61$, $G = 40$ or 100, 200 replicates per setting) with common random numbers. Except for the fixed-λ interval, every covariance is computed for the same REML fit of each replicate; these fits reproduce the stored fits of the main study to numerical precision. The intervals are the model-based interval ($\mathbf{V}_c$); CR1, CL2 with the $\mathbf{I} - \tilde{\mathbf{H}}_{ii}$ approximation, and CL2, all in the Bayesian form of Equation 9; CL2 in the frequentist form; CL2 with the critical value $t_{G-1,0.975}$ in place of $z_{0.975}$; and CL2 after a refit of the same data with the smoothing parameters fixed, per penalty, at the geometric mean of the setting's REML estimates. The fixed-λ refit removes the replicate-to-replicate variability of the smoothing parameters and also moves them to their typical level. Figure 8, Figure 9 and Figure 11 show the coverage, Figure 10 the paired differences; Supplement S7 lists the values.

Under dependence, the leverage adjustment adds 3.7 pp of coverage over CR1 at $G = 40$ and 1.2 pp at $G = 100$; the full block adds at most 0.4 pp in any setting over the $\mathbf{I} - \tilde{\mathbf{H}}_{ii}$ approximation (under dependence 0.1–0.4 pp), in line with $\mathbf{A}_i^{\mathrm{s}}\mathbf{R}_i\mathbf{A}_i^{\mathrm{s}} \preceq \mathbf{I}$. The allowance $\mathbf{V}_p - \mathbf{V}_e$ matters under independent errors (6.0 pp at $G = 40$), much less under dependence (1.4 pp).

With reselected smoothing parameters, the grid average of CL2's root-mean-square standard error divided by the empirical standard deviation of $\hat{\beta}$ under dependence is 0.92 at $G = 40$ and 0.97 at $G = 100$; with fixed smoothing parameters it is 1.05 and 1.06, and CL2 coverage of $\beta(s,t)$ rises from 0.929 to 0.945 at $G = 40$. Fixing the smoothing parameters, which Equation 9 conditions on, at their typical level closes most of the remaining shortfall at $G = 40$; the t_{G-1} critical value recovers 0.8 pp of it.

Fixed smoothing parameters also raise CL2 coverage of $\alpha(t)$ and $\gamma(t)$ under dependence (Figure 11): for binary responses with smooth errors at $G = 40$, CL2 coverage of $\alpha(t)$ rises from 0.885 to 0.925 and that of $\gamma(t)$ from 0.922 to 0.950. Under independent errors, the binary $\alpha(t)$ at $G = 40$ stays below nominal with fixed smoothing parameters (0.867).

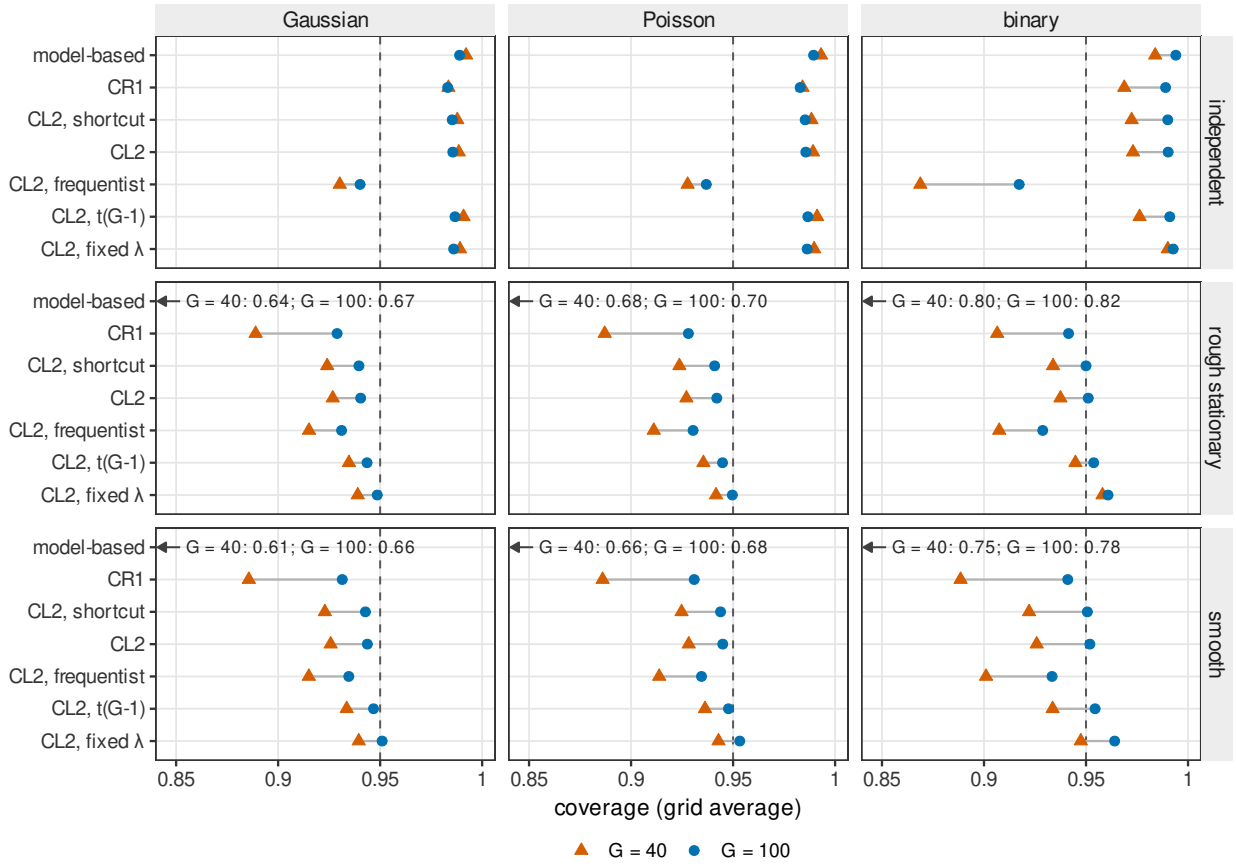


Figure 8: Covariance components, $\beta(s,t)$: grid-averaged coverage of nominal 95% pointwise intervals by family, error process and number of curves G, main factorial settings at the default signal level, 200 replicates. Model-based: $\mathbf{V}_c$; CR1, CL2, shortcut (CL2 with the $\mathbf{I}-\tilde{\mathbf{H}}_{ii}$ approximation) and CL2 in the Bayesian form; CL2, frequentist: frequentist form; CL2, t(G-1): CL2 with t critical value; CL2, fixed λ: CL2 after a refit with smoothing parameters fixed at the setting's geometric-mean REML values. Arrows at the left edge mark coverage below the axis range; the values are printed next to them. Dashed: 0.95.

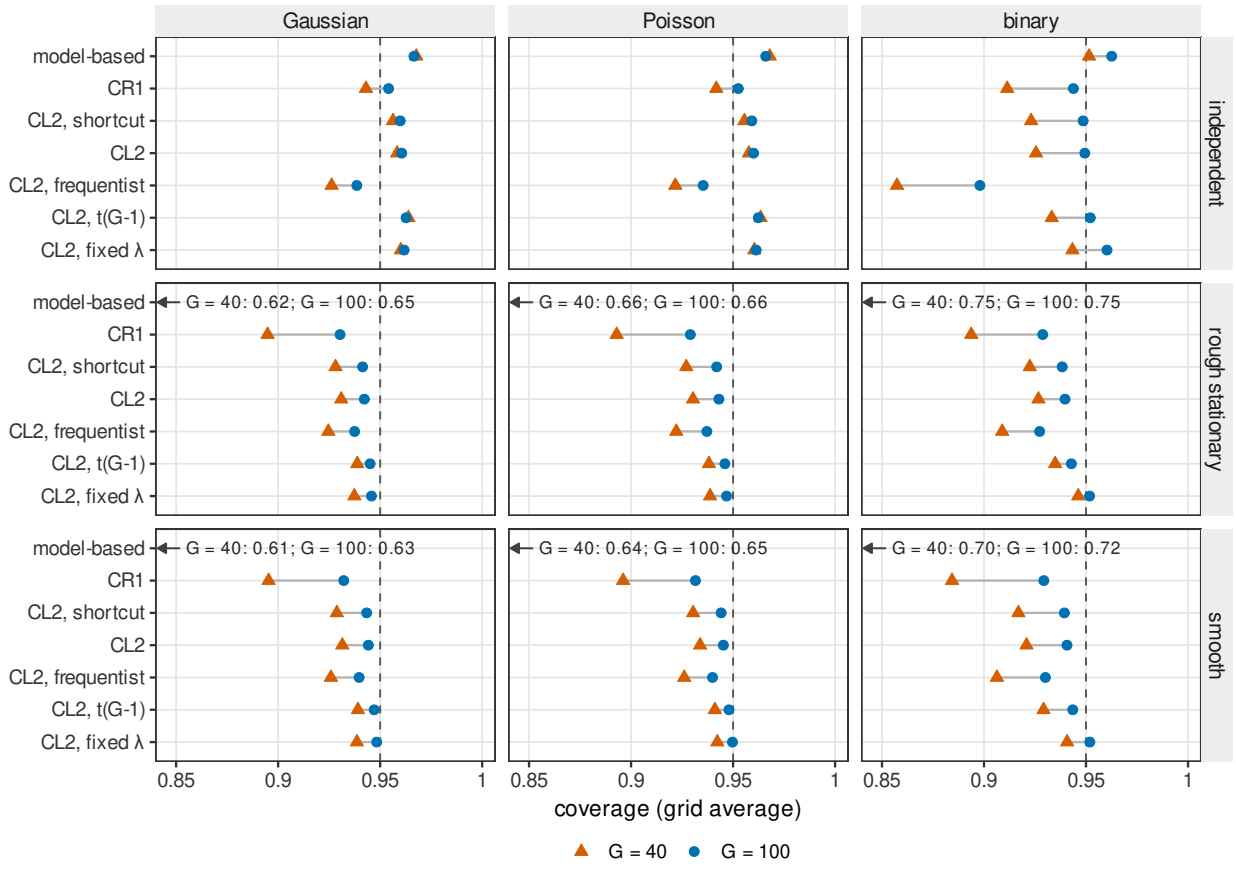


Figure 9: Covariance components, mean $\mathrm{E}(Y \mid X)$: grid-averaged coverage of nominal 95% pointwise intervals; intervals, settings and marks as in Figure 8.



Figure 10: Covariance components: paired coverage differences in percentage points, averaged over families and over the mean and $\beta(s,t)$, by error process and number of curves G. Each difference compares two intervals on the same replicates; unless stated, both are CL2 in the Bayesian form with the normal critical value. Shortcut: the $\mathbf{I}-\tilde{\mathbf{H}}_{ii}$ approximation. The SE of the averages cannot be recovered from the retained per-setting summaries, because the settings share replicates; the range of the per-setting paired Monte Carlo SEs is in Supplement S7.

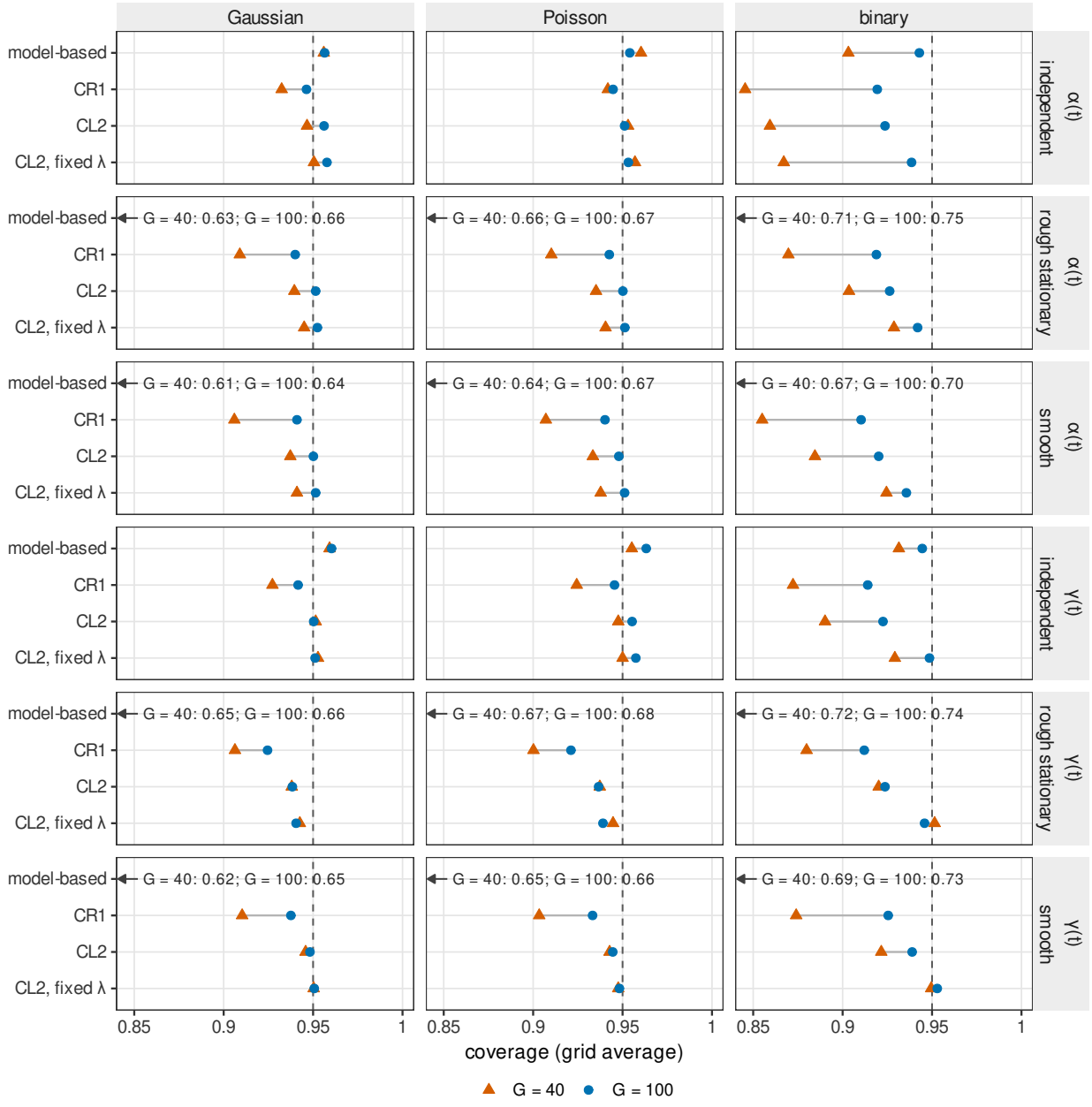


Figure 11: Covariance components, intercept $\alpha(t)$ and coefficient function $\gamma(t)$: grid-averaged coverage of nominal 95% pointwise intervals, settings and marks as in Figure 8; model-based ($\mathbf{V}_c$), CR1 and CL2 (Bayesian form), and CL2 after a refit with smoothing parameters fixed at the setting's geometric-mean REML values.

A.7 Satterthwaite degrees of freedom

For a contrast $\mathbf{a}^\top\theta$ the sandwich part of Equation 9 is a quadratic form in the whitened residuals, $\mathbf{a}^\top\mathbf{V}_p(\sum_i \mathbf{U}_i\mathbf{U}_i^\top)\mathbf{V}_p\mathbf{a} = \sum_i(\mathbf{q}_i^\top\mathbf{z}_i)^2$ with $\mathbf{q}_i = \mathbf{R}_i^{-1/2}\tilde{\mathbf{X}}_i\mathbf{V}_p\mathbf{a}$ (times $G/(G-1)$). Under the Gaussian working model with fixed smoothing parameters and no smoothing bias ($\mathbf{S}\theta = \mathbf{0}$, so that the residual mean $\mathbf{m}$ of Equation 11 vanishes), $\mathbf{z} = (\mathbf{I} - \tilde{\mathbf{H}})\varepsilon$ with independent errors. The quadratic form then has mean $\operatorname{tr}\Gamma$ and variance $2\operatorname{tr}(\Gamma^2)$, where the $G \times G$ matrix Γ has entries $\Gamma_{ih} = \mathbf{q}_i^\top[(\mathbf{I} - \tilde{\mathbf{H}})(\mathbf{I} - \tilde{\mathbf{H}})^\top]_{ih}\mathbf{q}_h$. Matching a scaled χ^2_ν gives the Satterthwaite degrees of freedom

$$\nu(\mathbf{a}) = \frac{(\operatorname{tr}\Gamma)^2}{\operatorname{tr}(\Gamma^2)} \tag{18}$$

(Bell and McCaffrey 2002; Pustejovsky and Tipton 2018), computed for every evaluation point; the interval uses $t_{\nu,0.975}$ in place of $z_{0.975}$. The Satterthwaite intervals evaluated here compute Equation 18 for the sandwich part only. A variant matches the moments of the full Bayesian CL2 variance, whose term $\mathbf{a}^\top(\mathbf{V}_p - \mathbf{V}_e)\mathbf{a}$ is fixed, which increases ν. Using only the diagonal of Γ gives practically the same ν.

With the floor inactive, $\Gamma_{ii} = \mathbf{q}_i^\top\mathbf{R}_i\mathbf{q}_i = \|\tilde{\mathbf{X}}_i\mathbf{V}_p\mathbf{a}\|^2$ and $\operatorname{tr}\Gamma = \mathbf{a}^\top\mathbf{V}_e\mathbf{a}$: Γ_{ii} is the contribution of curve i to the working-model variance of the contrast, and the diagonal version of ν, an upper bound, is $(\sum_i \Gamma_{ii})^2/\sum_i \Gamma_{ii}^2$, an effective number of curves behind the contrast (compare Imbens and Kolesár 2016). For the intercept, the curves contribute more equally and ν is larger. For a contrast that weights curves by a single covariate value or score x_i, as for $\gamma(t)$, Γ_{ii} is roughly proportional to x_i^2, and for Gaussian x_i, $(\sum_i x_i^2)^2/\sum_i x_i^4 \approx G/3$. Contrasts that involve several scores, as for $\beta(s,t)$ and fitted means, tend to give somewhat more. The degrees of freedom are smaller with heavy-tailed covariates or single curves of high leverage, which the covariates of our simulation do not include.

The Satterthwaite intervals use the REML fits and CL2 covariances of Appendix A.6 (the 18 main factorial settings at the default signal level, 200 replicates); Section 4.2, Section 5.2 and Supplement S1.2, S1.7 and S2.1 give them with 20 curves, in the extensions and on the plasmode data.

The median per-point degrees of freedom are 12–30 with $G = 40$ curves and 30–87 with $G = 100$; across both, that is 0.30–0.88 times $G - 1$.

Under dependent errors, Satterthwaite intervals cover the mean, $\beta(s,t)$ and $\gamma(t)$ at 0.95–0.96 with 40 curves (normal critical value: 0.92–0.94) and at 0.94–0.96 with 100 (0.93–0.95), at 1.10–1.12 and 1.04–1.05 times the width with the normal critical value (Figure 12).

Satterthwaite intervals do not repair the binary intercept: they cover 0.91–0.93 under dependence and 0.87–0.93 under independent errors. For this term, the Satterthwaite degrees of freedom are large (the intervals widen by a factor of only 1.01–1.05), so the reference distribution cannot make up for a standard error that is too small under independent errors or for the additional shift of the studentized error under dependence (Section 4.2, Supplement S1.4).

Under independent errors, Satterthwaite intervals overcover β further (0.98–0.99). The variant whose degrees of freedom include $\mathbf{V}_p - \mathbf{V}_e$ lies between t_{G-1} and the plain Satterthwaite reference.

Non-Gaussian responses. We use the same ν for all families (Section 2.3), although its derivation assumes Gaussian errors. Let the entries of ε be independent standardized score contributions with excess kurtosis κ_k (for Poisson responses, $\kappa_k = 1/\mu_k$), and let $\mathbf{Q}$ be the matrix with G columns whose ith column holds $\mathbf{q}_i$ in the rows of curve i and zeros elsewhere. With $\mathbf{B} = (\mathbf{I} - \tilde{\mathbf{H}})^\top\mathbf{Q}\mathbf{Q}^\top(\mathbf{I} - \tilde{\mathbf{H}})$, the quadratic form has, to first order, the same mean $\operatorname{tr}\Gamma$ but variance $2\operatorname{tr}(\Gamma^2) + \sum_k \kappa_k B_{kk}^2$. Positive excess kurtosis therefore makes

Equation 18 too large and the interval too narrow, and negative excess kurtosis does the reverse. The size of the extra term depends on $\sum_k B_{kk}^2 / \operatorname{tr}(\Gamma^2) \leq 1$. If $\tilde{\mathbf{H}}$ is ignored, this ratio is $1/D$ when all entries of the $\mathbf{q}_i$ have the same magnitude, so we expect little effect for contrasts that load on many grid points, unless κ_k is comparable to D. Sparse grids, localized contrasts and rare binary events are the less favorable cases. We did not compute this term or evaluate a family-dependent ν. Empirically, under dependent errors with 40 curves, Satterthwaite intervals cover the mean, β and γ at 0.95–0.96 for Gaussian, 0.95–0.96 for Poisson and 0.95 for binary responses.

B Setups of the two studies

B.1 Simulation design

The data follow the model in Equation 2 with $i = 1, \ldots, G$ curves. Responses are generated on a grid of 241 points in $t \in [0, 1]$ and subsampled to $D = 61$ (default) or 241 points; the functional covariate is observed on 51 points in $s \in [0, 1]$: $X_i(s) = \xi_{i0} + \sum_{k=1}^{29} \xi_{ik} \sqrt{2} \cos(k\pi s)$ with independent $\xi_{ik} \sim N(0, (k+1)^{-2})$. Its centered spectrum (first principal component 62% of the variance, four components 88%, rank 24 at 99.5%) matches the corpus-callosum profiles of the baseline DTI data in refund (67%, 86%, 26). The low-rank variant uses variances $(k + 1)^{-3}$ (rank 9). The scalar covariate is $v_i \sim N(0, 1)$. The smooth-effect model replaces the functional term by $f(x_i, t)$ with $x_i \sim U(0, 1)$.

Truths are fixed functions projected by least squares onto the fitted bases, so they are exactly representable: $\beta(s, t) = \cos(\pi s) \sin(\pi t) + (s - 1/2) \cos(2\pi t)$, $f(x, t) = 2(x - 1/2) \cos(\pi t) + t \sin(\pi x)$, $\gamma(t) = \sin(2\pi t)$. The rough truth adds $0.6 \sin(3\pi s) \sin(3\pi t)$ to β and f and $0.5 \sin(5\pi t)$ to γ. Truths are scaled so that the non-intercept signal $S = \int \operatorname{Var}_{X,v}\{\eta(t) - \alpha(t)\}\, dt$ is 1 (80% from the main term, 20% from γ) and multiplied by $\sqrt{S}$ per family. The intercept is a Gaussian bump plus a constant that sets the average response level (mean count 3, prevalence 0.2).

The fitted bases are cubic B-splines with first-order difference penalties (Eilers and Marx 1996; Wood 2017) (mgcv's `ps` basis with `m = c(2, 1)`), so constants along each direction are unpenalized; $k = 12$ along t for the intercept and γ; an 8×10 tensor product for β; 8×12 for f. The larger bases halve and quarter the knot spacing (about 100, 260 and 790 coefficients).

Within-curve errors are a zero-mean process with correlation kernel R, scaled for Gaussian responses and used as the latent layer of a Gaussian copula for the other families, $Y_{ij} = F_{ij}^{-1}\{\Phi(Z_{ij})\}$, so each marginal is exactly the fitted family and the working model is wrong only about the dependence. The kernels are: independent; rough stationary $0.8 \exp(-|t - t'|/0.089) + 0.2\, \mathbb{1}\{t = t'\}$; smooth: the residual covariance of a REML fit to the DTI data (24 eigenfunctions plus 6% white noise, as a correlation kernel); exact AR(1) $\exp(-|t - t'|/0.071)$; sign-changing $0.8 \exp(-|t - t'|/0.3) \cos(2\pi|t - t'|/0.4) + 0.2\, \mathbb{1}\{t = t'\}$; heteroskedastic versions of the smooth kernel with the DTI variance profile along t, a subject scale $\exp(0.5v_i - 0.25)$, or both. Figure 13 shows the homoskedastic kernels.

Under misregistration, $Y_i(t_j) = \eta_i(h_i(t_j)) + \varepsilon_i(h_i(t_j))$ with $h_i(t) = t + a_i \sin(\pi t)/\pi$, $a_i \sim N(0, 0.314^2)$ truncated at $|a_i| \leq 0.9$, and errors are independent. Gaussian misregistration settings score the warp-averaged coefficient functions, GLM settings only the exact observed-clock mean.

The signal is $R^2 \in \{0.2, 0.5, 0.8\}$ for Gaussian responses and $\sqrt{S} \in \{0.5, 1\}$ on the link scale for Poisson and binary responses. The extended families are run at the default signal: scaled t (4 df) as Gaussian at $R^2 = 0.5$; beta ($\phi = 20$, mean 0.3) and negative binomial (size 5, mean 3) at $\sqrt{S} = 0.5$.

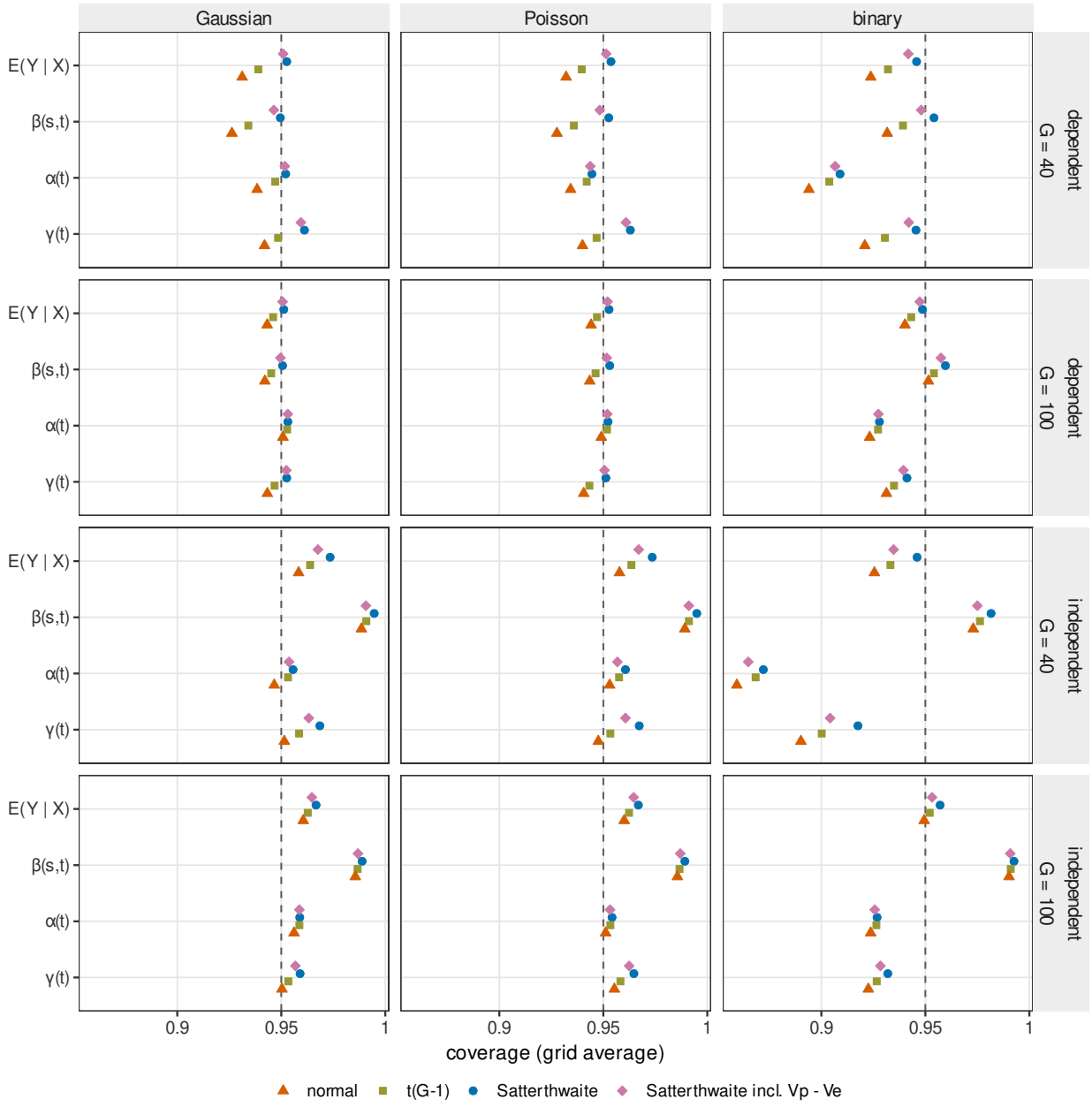


Figure 12: Critical values for REML + CL2: grid-averaged coverage by dependence, number of curves G, family and estimand; means over the error processes of the main factorial at the default signal level, 200 replicates. Satterthwaite: per-point degrees of freedom for the sandwich part; incl. Vp - Ve: degrees of freedom for the full Bayesian CL2 variance. Dashed: 0.95. The 5% quantile of pointwise coverage and the width relative to the normal critical value are in Supplement S7.

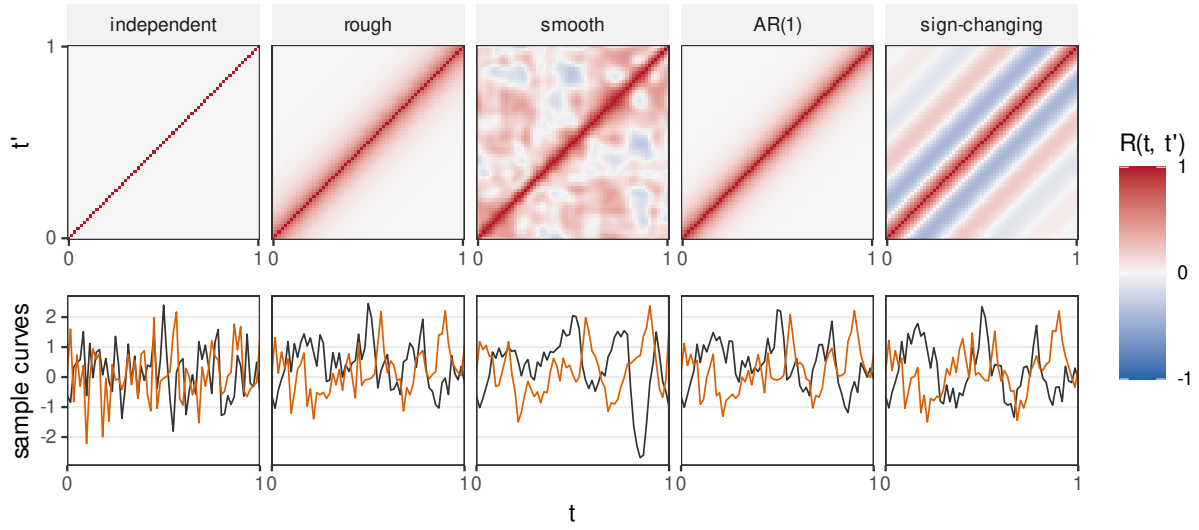


Figure 13: Error processes of the simulation: correlation kernels $R(t, t')$ (top) and two sample curves of the latent Gaussian process (bottom, same draws of standard normals across processes), on the default grid of $D = 61$ points. The main factorial uses the first three.

The 20-curve block uses the first 20 curves of each replicate of the main factorial, so it shares its draws with the 40- and 100-curve settings. The zero-surface block sets $\beta \equiv 0$ in the smooth-truth setting at the default signal, for Gaussian and binary responses with independent and smooth errors and 40 and 100 curves; γ, the intercept bump and the Gaussian error variance are those of the non-zero setting, and the intercept constant of binary responses is reset to keep the prevalence at 0.2.

All random draws depend only on the replicate index, so contrasts between settings and intervals are paired. Each setting has 200 replicates; none failed. Concurvity of the fitted terms (mgcv's worst case) is at most 0.25. The next table lists the blocks of settings: the main factorial (family × error process × G × signal) and extensions that each change one factor.

Table 3: Blocks of the simulation design. Error processes: indep. independent, rough rough stationary, smooth from the DTI residuals, AR(1) exact AR(1), sign-chg. sign-changing, var(t)/var(z)/var(t,z) heteroskedastic versions of smooth, misreg. misregistration. Truth: smooth, rough or zero. Signal: low, default, high. Bases: default, 2× or 4× finer knots.

block	settings	families	errors	G	signal	truth	D	model	covariate	basis
main factorial	42	Gaussian, Poisson, binary	indep., rough, smooth	40, 100	low, default, high	smooth	61	functional	rich	default
main factorial, G = 20	21	Gaussian, Poisson, binary	indep., rough, smooth	20	low, default, high	smooth	61	functional	rich	default
misregistration	12	Gaussian, Poisson, binary	misreg.	40, 100	high	smooth, rough	61	functional	rich	default
dense grid	9	Gaussian, Poisson, binary	indep., rough, smooth	100	default	smooth	241	functional	rich	default
rough truth	14	Gaussian, Poisson, binary	indep., smooth	100	low, default, high	rough	61	functional	rich	default
zero surface	8	Gaussian, binary	indep., smooth	40, 100	default	zero	61	functional	rich	default
smooth effect	9	Gaussian, Poisson, binary	indep., rough, smooth	100	default	smooth	61	smooth effect	rich	default
extended families	8	scaled t, beta, negative binomial, negative binomial (fitted as Poisson)	indep., smooth	100	default	smooth	61	functional	rich	default
exact AR(1)	1	Gaussian	AR(1)	100	default	smooth	61	functional	rich	default
sign-changing	3	Gaussian, Poisson, binary	sign-chg.	100	default	smooth	61	functional	rich	default
misreg. (AR(1) fit)	1	Gaussian	misreg.	100	high	rough	61	functional	rich	default
heteroskedastic	6	Gaussian	var(t), var(z), var(t,z)	40, 100	default	smooth	61	functional	rich	default
low-rank covariate	6	Gaussian, Poisson, binary	indep., smooth	100	default	smooth	61	functional	low-rank	default
larger bases	12	Gaussian, Poisson, binary	indep., smooth	100	default	smooth	61	functional	rich	2× finer knots, 4× finer knots

Each replicate is fitted by REML and curve-blocked NCV (mgcv `gam`); by pointwise (non-blocked) NCV in the Gaussian settings at $G = 100$ with independent, smooth and sign-changing errors; and by the AR(1) working model in the Gaussian settings at $G = 100$ with dependent errors. The AR(1) model is fitted with mgcv's `bam()` by fast REML with AR(1) errors that restart at the first grid point of each curve. Its correlation ρ is profiled over the grid $\{0, 0.1, \ldots, 0.9, 0.95, 0.99\}$ by the fast-REML criterion, which includes the log-determinant of the whitening transform; the best grid value is then refined by a one-dimensional search between its two neighboring grid values. Its intervals are model-based from that fit. The plasmode study uses the same fit.

Some NCV fits were flagged as not converged, meaning that mgcv's search over the smoothing parameters ended without meeting its convergence test: 88% of scaled-t fits under smooth errors and 68% under independent errors, 1.0% of beta fits, and fewer than 0.1% otherwise (settings with at least 40 curves and a non-zero surface; at most 2.5% per combination in the plasmode study). No REML fit in these settings was flagged, so the study says nothing about intervals from non-converged REML fits.

We refitted flagged scaled-t fits from the REML smoothing parameters and from ten times those values. For the first five flagged fits, all with independent errors, the refits reach the same criterion within 8×10^{-5} and the same surface within 6×10^{-4}. Of 20 randomly chosen flagged fits with smooth errors, 17 reach the same criterion within 0.005, the same surface within 0.004 and the same CL2 standard errors within 0.3%, consistent with a flat optimum; in an 18th the study fit's criterion is 2.2 above the refits', with the surface within 0.04. In the other two, a refit returns a materially different fit, its surface moved by about twice the root-mean-square size of the true surface (1.8). In one, the study fit has the worse criterion, by 17, and the standard errors differ by up to 80%; in the other, a refit is worse by 10, with standard errors up to 403% larger. 38 of the 40 refits were flagged too. A flag thus does not separate these fits from the stable ones.

Under smooth errors, in the 176 replicates with a flagged NCV fit, the NCV estimate of β has median relative error 0.13 and NCV + CL2 covers 0.938 (Monte Carlo SE 0.004), against 0.12 and 0.959 (0.008) in the 24 other replicates; this difference of 2.1 pp is the largest coverage gap of NCV + CL2 over estimands and error processes. The model-based NCV intervals cover 0.697 and 0.742, REML + CL2 0.943 and 0.951 in the same two groups, so part of the gap may reflect harder data. The flag is not randomly assigned and the comparison is descriptive.

Low-smoothing NCV fits (Section 4.1) were examined in the main factorial and in the settings with denser grids, other families, rough truths, AR(1) errors and misregistration (86 settings with 40 or 100 curves), with the tail defined on the first surface penalty. Under dependence, apart from misregistration, cutoffs of e^4 and e^2 give 4% and 9% of NCV fits, and applying the $e^3 \approx 20$ cutoff to either penalty gives 7.4%. Under misregistration, NCV's smoothing parameters are the more variable ones, with up to 24% of fits below the e^3 cutoff (Poisson). Under dependence, apart from misregistration, the share of low-smoothing fits is 4.0–7.9% across families and error processes, and these fits have 1.4 times their setting's median surface EDF, still fewer than REML's on all of them. REML also does worse on these replicates: its error is 1.6 times its median and its CL2 intervals cover 0.858 (NCV + CL2: 0.831). On the remaining replicates, REML's mean squared error is 8.9 rather than 6.9 times NCV's (geometric means over settings). The converse does not hold: on the 17% of replicates in REML's own lower tail, NCV's error stays near its median (1.05 times) and its coverage is 0.6 pp lower than on the other replicates.

B.2 Plasmode datasets

A plasmode simulation (Franklin et al. 2014) keeps the real covariates and real error structure of a dataset and replaces only the true regression function by a known one.

ECG strain (independent curves; myocardial-strain study, first beat per subject; non-public; aggregates only). 78 curves; response: global longitudinal strain, 4-chamber view (%), one cardiac cycle normalized to [0, 1], native grid about 52 frames per beat (29–83) (fitted: interpolated to 61 points); functional covariate: simultaneously recorded ECG of the same beat (digitized pixel intensity, shape usable, amplitude up to scale), 51 points; scalar covariate: age (standardized). Preprocessing: cycle time normalized to the R–R interval; subjects with at least two beats (the second beat supplies the beat-to-beat difference curves). REML fit: R^2 0.47, surface EDF 59.6 (NCV fit: 44.0); residual design effect 19.9, lag-one correlation 0.96, 15% negative correlations, 4 principal components for 90% of the residual variance, covariate rank 15.

AF trial (independent curves; atrial-fibrillation trial, patients randomized to digoxin or a beta-blocker; non-public; aggregates only). 118 curves; response: left-atrial strain, 4-chamber view, at day 30 (%), one cycle on [0, 1], native grid 30 points per cycle (fitted: used as is); functional covariate: the same strain curve at day 0, 51 points; scalar covariate: randomized treatment (digoxin = 1). Preprocessing: none beyond the cycle normalization; a baseline-adjusted functional ANCOVA. REML fit: R^2 0.19, surface EDF 33.2 (NCV fit: 25.3); residual design effect 10.9, lag-one correlation 0.89, 20% negative correlations, 3 principal components for 90% of the residual variance, covariate rank 8.

running (independent curves; Fukuchi, Fukuchi & Duarte 2017 (PeerJ 5:e3298); one row per person, first condition; public (tidyfundata, MIT; original data public)). 90 curves; response: sagittal knee moment over normalized stance, native grid 101 points (fitted: interpolated to 61 points); functional covariate: sagittal knee angle over the same stance, 51 points; scalar covariate: body mass (standardized). Preprocessing: one row per person (post-intervention rows dropped); speed and backpack conditions differ between persons and stay in the residuals. REML fit: R^2 0.46, surface EDF 53.7 (NCV fit: 54.8); residual design effect 33.3, lag-one correlation 0.99, 12% negative correlations, 2 principal components for 90% of the residual variance, covariate rank 7.

DTI (independent curves; refund::DTI, baseline visit (Goldsmith et al. 2011): multiple-sclerosis patients and controls; public (refund, GPL ≥ 2)). 92 curves; response: fractional anisotropy along the right corticospinal tract, native grid 55 points (fitted: used as is); functional covariate: fractional anisotropy along the corpus callosum, 51 points; scalar covariate: case status (MS = 1). Preprocessing: complete cases of both tracts. REML fit: R^2 0.21, surface EDF 53.8 (NCV fit: 35.1); residual design effect 9.9, lag-one correlation 0.81, 19% negative correlations, 16 principal components for 90% of the residual variance, covariate rank 24.

gait (independent curves; Van Criekinge et al. 2023 (Sci Data 10:852; figshare 24192489), able-bodied adults walking barefoot at preferred speed; public (CC0)). 138 curves; response: sagittal knee moment over one stride, one mean stride per subject, native grid 1001 points (0.1% of the stride) (fitted: every 10th point (101)); functional covariate: sagittal knee angle over the stride, 51 points; scalar covariate: age (standardized; from the subject-characteristics file). Preprocessing: the authors' stride normalization and averaging over strides (within-subject stride variability is averaged out). REML fit: R^2 0.40, surface EDF 68.7 (NCV fit: 60.8); residual design effect 37.0, lag-one correlation 0.98, 33% negative correlations, 6 principal components for 90% of the residual variance, covariate rank 8.

ECG 8-lead (independent curves; roahd::mfD_healthy and mfD_LBBB (Ieva et al.), 50 healthy and 50 left-bundle-branch-block subjects; public (roahd, GPL-3)). 100 curves; response: ECG lead V5 (μV) over one beat, native grid 1024 points at 1 kHz (fitted: interpolated to 101 points); functional covariate: ECG lead I, 51 points; scalar covariate: LBBB indicator. Preprocessing: registered and smoothed by the package authors. REML fit: R^2 0.21, surface EDF 39.4 (NCV fit: 11.2); residual design effect 6.2, lag-one correlation 0.94, 32% negative correlations, 4 principal components for 90% of the residual variance, covariate rank 15.

ocean (dependent curves; Hawaii Ocean Time-series profiles (FRegSigCom::ocean), monthly casts; public (FRegSigCom, GPL ≥ 2; HOT data public)). 116 curves; response: dissolved oxygen over depth, native grid

101 points (fitted: used as is); functional covariate: temperature over depth, 51 points; scalar covariate: cast order (standardized). Preprocessing: none; casts kept in time order. REML fit: R^2 0.36, surface EDF 59.2 (NCV fit: 49.8); residual design effect 39.7, lag-one correlation 0.99, 22% negative correlations, 5 principal components for 90% of the residual variance, covariate rank 6.

weather (dependent curves; AEMET open data via fda.usc::aemet, 73 Spanish stations, daily means 1980–2009; public (fda.usc, GPL-2)). 73 curves; response: log precipitation over the year, native grid 365 days (fitted: every 5th day (73)); functional covariate: daily mean temperature over the year, 51 points; scalar covariate: altitude (standardized). Preprocessing: none beyond the day thinning. REML fit: R^2 0.41, surface EDF 29.5 (NCV fit: 38.6); residual design effect 29.1, lag-one correlation 0.50, 1% negative correlations, 19 principal components for 90% of the residual variance, covariate rank 4.

electricity (dependent curves; Adelaide electricity demand (fds::mondaydemand, fds::mondaytempairport; Magnano, Boland & Hyndman 2008), every 5th Monday 1997–2007; public (fds, GPL-3)). 102 curves; response: half-hourly electricity demand over the day, native grid 48 half-hours (fitted: used as is); functional covariate: airport temperature over the same day, 51 points; scalar covariate: week index (standardized; absorbs the linear trend). Preprocessing: every 5th Monday (thins, not removes, the week-to-week dependence; the annual cycle stays). REML fit: R^2 0.31, surface EDF 33.2 (NCV fit: 10.0); residual design effect 35.8, lag-one correlation 0.98, 0% negative correlations, 2 principal components for 90% of the residual variance, covariate rank 11.

The ECG strain beats were recorded with 29 to 83 frames and interpolated to 61 grid points; interpolation adds smoothness within a curve, so part of the residual dependence of that dataset, and of its design effect, comes from the interpolation.

Six datasets have independent source curves. The other three have serially (ocean, electricity) or spatially (weather) dependent source curves.

The residual *design effect* of a dataset is $\mathbf{1}^\top \Sigma \mathbf{1} / \operatorname{tr} \Sigma$ for the covariance Σ of its REML residual curves: the factor by which the variance of a grid mean exceeds its working-independence value.

Table 4: Fitted models: bases (intercept, γ; surface 8×10), R^2 of the REML fit, surface and total EDF of the three fits used as truths (REML / NCV / midpoint), and whether the midpoint fit lies between the two for every term.

dataset	intercept basis k	gamma(t) basis k	surface basis (s × t)	R^2	surface EDF REML / NCV / midpoint	total EDF REML / NCV / midpoint	midpoint interior
ECG strain	12	12	8 × 10	0.469	59.6 / 44.0 / 52.9	77.1 / 54.1 / 63.4	FALSE
AF trial	12	12	8 × 10	0.192	33.2 / 25.3 / 29.5	44.7 / 32.8 / 38.9	TRUE
running	12	12	8 × 10	0.458	53.7 / 54.8 / 60.4	68.0 / 65.9 / 73.1	FALSE
DTI	12	12	8 × 10	0.210	53.8 / 35.1 / 47.0	68.5 / 47.9 / 60.7	TRUE
gait	12	12	8 × 10	0.397	68.7 / 60.8 / 65.3	89.0 / 80.1 / 85.2	FALSE
ECG 8-lead	40	30	8 × 10	0.211	39.4 / 11.2 / 23.9	103.2 / 77.0 / 88.9	TRUE
ocean	12	12	8 × 10	0.362	59.2 / 49.8 / 60.3	74.3 / 58.7 / 72.1	FALSE
weather	12	12	8 × 10	0.407	29.5 / 38.6 / 41.9	45.4 / 60.8 / 61.5	FALSE
electricity	12	12	8 × 10	0.312	33.2 / 10.0 / 23.9	45.8 / 22.0 / 35.7	FALSE

Table 5: Residual dependence and covariate diagnostics (REML residual curves): design effect DE and DE/D, residual correlation at one grid step and at 10% and 25% of the domain, share of negative correlations and the most negative one, max/min SD along t, principal components for 90% of the residual variance, centered covariate rank, share of response variance removed by registration.

dataset	DE	DE/D	lag 1 step	lag 10%	lag 25%	neg. corr. share	min r	sd ratio	FPC90	rank X	phase share
ECG strain	19.88	0.326	0.962	0.620	0.208	0.146	-0.351	24.53	4	15	0.63
AF trial	10.93	0.364	0.889	0.684	0.359	0.195	-0.269	29.78	3	8	0.43
running	33.31	0.546	0.994	0.871	0.594	0.119	-0.095	4.63	2	7	0.30
DTI	9.91	0.180	0.811	0.233	0.106	0.193	-0.340	2.31	16	24	0.22
gait	36.96	0.366	0.978	0.483	0.258	0.327	-0.535	6.15	6	8	0.03
ECG 8-lead	6.19	0.061	0.938	0.363	0.078	0.324	-0.681	42.26	4	15	0.33
ocean	39.74	0.393	0.987	0.738	0.432	0.217	-0.268	2.86	5	6	0.00
weather	29.12	0.399	0.503	0.489	0.393	0.006	-0.101	2.22	19	4	0.06
electricity	35.77	0.745	0.981	0.894	0.822	0.000	0.308	3.07	2	11	0.00

Table 6: Datasets with dependent source curves: block construction for the block flips and block-wise NCV; residual-score correlation within and between blocks.

dataset	block rule	blocks	block size	alt. blocks	score corr. within	score corr. between	between pairs
ocean	3 consecutive monthly casts (cost check; the autocorrelation rule gives single casts)	39	2–3		0.045	-0.100	38
weather	complete-linkage clusters of station distance cut at 250 km (mismatched partition: 150 km)	16	1–8	25	0.548	0.356	276
electricity	2 consecutive frame curves (10 weeks)	51	2		0.214	0.011	50

B.3 Construction of the plasmode data

Each dataset's model, $Y_i(t) = \alpha(t) + \int X_i(s)\beta(s,t)\,ds + \nu_i\gamma(t) + \varepsilon_i(t)$ (Gaussian), is fitted by REML and by curve-blocked NCV. The fitted means of three fits serve as truths: the REML fit (*REML-fitted truth*), the NCV fit (*NCV-fitted truth*) and a refit with every smoothing parameter at the geometric mean of the two fits' values (*midpoint truth*). Replicate data are these fitted means plus each subject's own residual curve from the REML or the NCV fit, multiplied by an independent random sign per curve. Sign flipping keeps each residual curve's within-curve covariance and makes the curves independent across subjects. For the ECG strain data, heartbeat difference curves $(Y_{i1} - Y_{i2})/\sqrt{2}$ of two beats per subject are a further error source, not shaped by any fit, used on their own scale or rescaled to the average variance of the REML residuals.

Each combination of dataset, truth and residual source is run with all subjects and with 40 randomly chosen subjects (200 replicates each). The three datasets with dependent source curves add block flips (one sign per block of neighboring curves) and block-wise NCV and CL2. The mean is evaluated at the covariates of a fixed set of 50 subjects of the dataset, at the native grid points. There are 130 combinations in all.

Table 7: Choice of the fixed set of 50 subjects for the mean: the rule used, and the balance check of the first 50 subjects against the remaining subjects (one-sided test at 5%, permutation p-value, standardized mean differences of the scalar covariate and the first three covariate scores). Where this check failed, a seeded random sample of 50 subjects was used; the balance of that sample was not recorded.

dataset	G	cohort	one-sided at 5%	permutation p	SMD z	SMD PC1	SMD PC2	SMD PC3
ECG strain	78	first 50 rows	FALSE	0.302	0.313	0.021	-0.180	-0.402
AF trial	118	seeded sample of 50	TRUE	0.000	1.788	0.046	-0.231	0.317
running	90	seeded sample of 50	TRUE	0.000	-0.331	-0.398	0.916	-0.008
DTI	92	seeded sample of 50	TRUE	0.000	-1.149	-0.408	-0.258	-0.036
gait	138	seeded sample of 50	TRUE	0.000	1.668	0.118	0.299	0.377
ECG 8-lead	100	seeded sample of 50	TRUE	0.000	-1.990	-1.344	0.235	0.274
ocean	116	seeded sample of 50	TRUE	0.000	-1.725	-0.255	-0.346	0.109
weather	73	first 50 rows	FALSE	0.411	-0.173	0.130	0.225	0.378
electricity	102	seeded sample of 50	TRUE	0.000	-1.724	0.029	-0.091	-0.297

C Computing time

Times are wall-clock seconds on one core. They come from four sources: (A) a benchmark on one local core with Gaussian data, $G = 100$ curves, $D = 61$ grid points and the default bases (90 replicates over the dependent error processes of the simulation); (B) the comparator runs on a computing cluster, Gaussian main-factorial settings and plasmode datasets with independent source curves, $G = 40$ to 138; (C) the smoothing-parameter-term runs on the main-factorial settings at the default signal ($G = 40$ and 100); (D) the same terms on the plasmode data. Figure 14 shows the median, the quartiles, the 90% quantile and the maximum per method; Supplement S7 lists them.

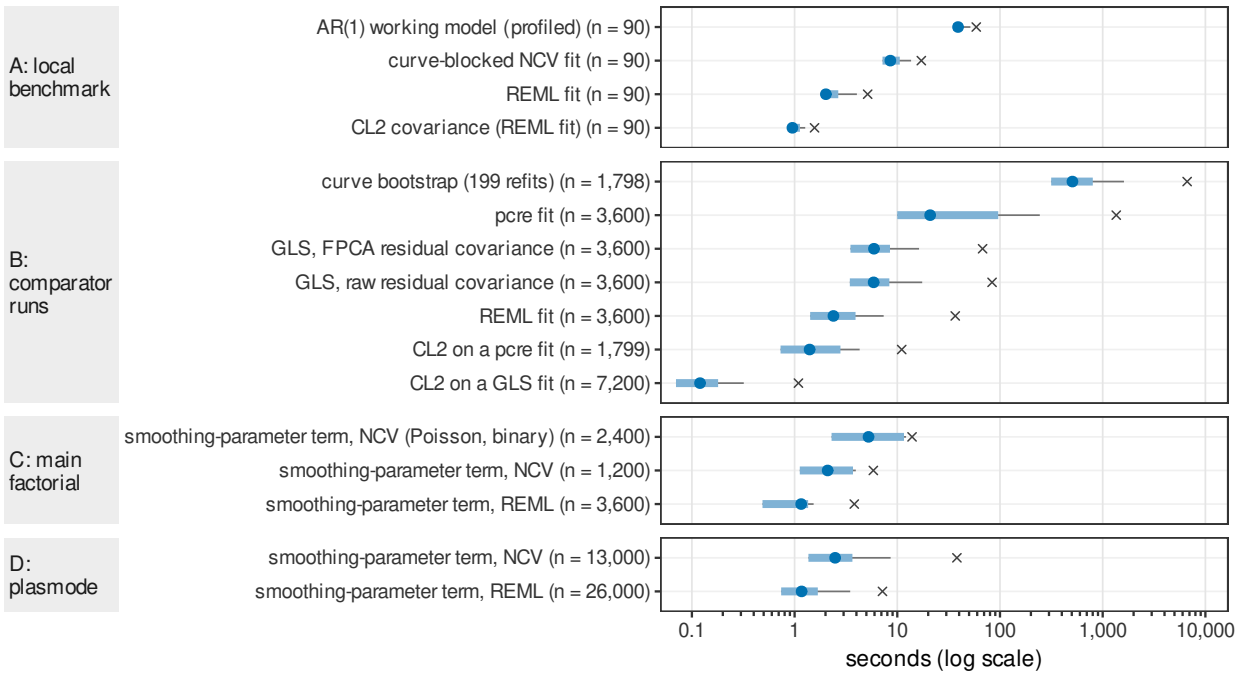


Figure 14: Computing time in seconds per fit or per covariance, by method and data source (A: local benchmark, Gaussian, G = 100, D = 61; B: comparator runs, Gaussian synthetic settings and plasmode datasets with independent source curves; C: main-factorial settings; D: plasmode data); n: number of timed fits or covariances; log scale. Point: median; thick bar: quartiles; thin bar: up to the 90% quantile; cross: maximum. CL2 and the smoothing-parameter terms are computed on top of an existing fit. Times for pcre and GLS include the REML pilot fit and, where used, the FPCA of its residual curves.

A curve-blocked NCV fit costs about 4.2 times a REML fit; the profiled AR(1) model about 19 times, because it refits the model at each value of ρ. The CL2 covariance of a REML fit adds a median 0.95 s, its Satterthwaite degrees of freedom add a median 1.8 s per replicate for all four estimands in the main factorial, 2.1 s in the other synthetic settings and 1.7 s on the plasmode data (cluster nodes).

Times for GLS and pcre include the REML pilot fit and the FPCA of its residual curves (the raw GLS covariance needs no FPCA). GLS then takes a median of 5.9 s; pcre's time grows quickly with the number of curves and grid points (median 21 s, 90% quantile 244 s, maximum 1356 s). CL2 on these fits adds a median of 0.12 s (GLS) and 1.4 s (pcre). The curve bootstrap with 199 refits takes a median of 507 s and up to 6645 s per dataset. The smoothing-parameter uncertainty terms (curve scores, Hessian and infinitesimal jackknife) add a median of 1.2 s to a REML fit and 2.1 s to a Gaussian NCV fit. The one-step bias correction of the NCV estimate needs one linear solve with $p \times p$ matrices and was not timed separately.